

Research Portfolio

Evidence of Extraordinary Ability in Cybersecurity Research

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Cybersecurity — Vulnerability Management & Anomaly Detection

Portfolio date: May 31, 2026
Papers included: 4 original research papers
Venues targeted: IEEE, USENIX, ACM (peer-reviewed)
Data policy: All data synthetic and seeded;
no employer or production data used
Code policy: All code and results publicly released

Contents of This Portfolio:

1. **Exhibit A** — Paper 1: VulnPrio — Vulnerability Prioritization Benchmark
2. **Exhibit B** — Paper 2: CalibScore — Failure-Aware Calibration Gate
3. **Exhibit C** — Paper 3: HygieneBench — Cyber-Hygiene Anomaly Detection Benchmark
4. **Exhibit D** — Paper 4: HygienePrio — Hygiene-Augmented Exploit-Weighted Scorer

Each exhibit includes: (1) a one-page layperson significance statement; (2) a technical abstract; and (3) the full paper draft.

EXHIBIT A

Paper 1: **VulnPrio**

Context-Aware Vulnerability Prioritization for Government Endpoint Fleets: Integrating Exploit Intelligence, Asset Criticality, and Endpoint Telemetry

Harshavardhan Malla — Independent Researcher

Target venue: IEEE/USENIX Security Workshop on Measurement and Practice

Layperson Summary:

Government agencies receive thousands of new software vulnerability reports each month but can only fix a fraction of them before the next wave arrives. The standard approach — ranking by “severity score” (CVSS) — does not account for whether a vulnerability is actually being exploited in the wild, how critical the affected computer is, or how complex the patch is to apply. This paper introduces **VulnPrio**, a mathematical framework that combines seven signals (exploit probability, known exploitation status, severity, asset criticality, network exposure, urgency, and remediation difficulty) into a single priority score, and embeds that score in a scheduling system that automatically handles regulatory deadlines and records every decision in a tamper-evident log.

The key finding is deliberately honest: under the current research conditions, VulnPrio performs no better than simpler approaches — and the paper says so explicitly. This “null result” reported with full statistical rigor is the contribution: it establishes a reproducible testing framework that any future system can be compared against, and proves that data-driven superiority claims require real calibrated data — not placeholder weights.

Field Impact:

- First benchmark combining EPSS signals, capacity constraints, KEV-deadline overrides, and hash-chained audit logs in one reproducible pipeline
- 390 hash-chained audit logs, all verifying without error — a tamper-evident decision record mechanism suitable for regulatory environments
- Honest null-result reporting with full statistical apparatus (Wilcoxon + Holm + BCa bootstrap) across 30 seeds and 13 strategies
- Open code and frozen results released for community replication

Technical Abstract:

Vulnerability remediation in large government endpoint fleets is constrained by finite patch capacity, mandatory regulatory timelines, and the daily flood of newly published CVEs. We present **VulnPrio**, a seven-feature linear prioritization framework integrat-

ing EPSS exploit probability, KEV status, CVSS base score, asset criticality, network exposure, urgency, and remediation complexity. The framework is embedded in a five-phase capacity-constrained scheduler with KEV-deadline overrides and append-only SHA-256 hash-chained audit logs. Evaluated against 12 competing strategies across 30 seeds on a fully synthetic fleet dataset, all 13 strategies are *statistically indistinguishable* on the Expected Exploited Host-Days (EHD) metric (inter-strategy spread $< 0.5\%$; all 78 pairwise Wilcoxon+Holm tests: adjusted $p > 0.05$). We report this null result honestly: the contribution is a reproducible, falsifiable, audit-evidence-producing benchmark, not a performance claim. All 390 audit logs hash-verify correctly.

EXHIBIT B

Paper 2: CalibScore

When Calibration Fails: A Failure-Aware Public-Feed Gate for Vulnerability Prioritization Under Sparse Exploit Labels

Harshavardhan Malla — Independent Researcher

Target venues: USENIX CSET (primary); LASER Workshop; ACM DTRAP

Layperson Summary:

Paper 1 uses placeholder (estimated) weights in its scoring formula because no one had measured whether real public data could calibrate them. Paper 2 asks that question directly: can we use publicly available exploit records to find better weights for a realistic government fleet? The answer is no — and this paper explains precisely why, with a formal measurement.

The paper introduces a **failure-aware gate**: a step-by-step procedure for deciding whether you have enough data to calibrate a model before you try. The gate was written down (pre-registered) before any data was collected, so there was no possibility of changing the rules after seeing the results. When applied to a realistic 31-product government fleet, the gate found only 7 confirmed exploitation events over two years — far below the minimum of 20 needed to safely fit a calibration model. The gate said “PIVOT — do not calibrate,” and the paper reports exactly that outcome as its primary finding.

Field Impact:

- First paper to propose a formal “feasibility gate” before attempting public-feed calibration of vulnerability prioritization weights
- Pre-registration of analysis plan (8 stop rules) before any API queries were executed — a clinical-trial-grade methodological standard applied to cybersecurity research
- Quantified data-density boundary: provides a scaling formula ($E[\text{positives}] \approx p \cdot r \cdot m \cdot T$) for practitioners to estimate feasibility before committing to a calibration study
- BCa confidence intervals so wide (Top-10 coverage: [0.00, 0.71]) that they honestly communicate irreducible uncertainty — no false precision

Technical Abstract:

Exploit-occurrence labels from the CISA KEV catalog are sparse in bounded product catalogs. We contribute a pre-registered, multi-stage gate methodology for assessing whether public-feed signals (EPSS v3, NVD, KEV) are dense enough to calibrate vulnerability-prioritization weights without label leakage. The gate comprises: chunked API acquisition with checkpointing; an 18-window multi- t_0 aggregation that maximizes observable distinct positive CVEs; and a three-tier decision rule (PROCEED ≥ 50 , CAUTION 20–49, PIVOT < 20) backed by a pre-registered stop-rule registry. Applied to 2,688 catalog-matched CVEs over the EPSS v3 era (2023-03-07 to 2025-

03-16), we observe only **7 unique positive CVEs** across 18 windows — triggering a gate decision of **PIVOT**: calibration is not attempted. The gate PIVOT is the primary contribution. A pre-registered sensitivity sweep of six design-prior weight vectors produces descriptive statistics with wide BCa CIs, confirming that no vector can be distinguished under this sparsity regime.

EXHIBIT C

Paper 3: HygieneBench

HygieneBench: A Reproducible Synthetic Benchmark for Cyber-Hygiene Anomaly Detection Across Identity, Endpoint, and Patch Telemetry

Harshavardhan Malla — Independent Researcher

Target venues: ACM CCS AISEC Workshop (primary); IEEE S&P Workshop

Layperson Summary:

Good cyber security depends on basic hygiene: making sure old accounts are closed, computer agents are running, patches are applied on time, and data is being collected reliably. When these hygiene checks fail, the conditions exist for attackers to exploit dormant accounts, unmonitored computers, or unpatched vulnerabilities. Detecting hygiene failures automatically is therefore a fundamental security task — but there is no public testing dataset for this problem. Researchers developing AI tools for hygiene monitoring have had no standard way to compare their methods.

HygieneBench fills this gap. It generates realistic (but fully synthetic) organizational data spanning employee accounts, computers, software patches, vulnerability records, and data freshness logs. It includes 12 types of hygiene anomalies and 7 detection tasks, and it tests 8 different detection methods across 5 conditions (including “stale data” conditions that simulate real-world data pipeline failures). The central finding is surprising: **simple rule-based detection** outperforms sophisticated machine learning on 86.2% of tested configurations. Only two tasks (detecting unusual account group changes, and detecting clusters of unpatched critical vulnerabilities) consistently benefit from machine learning.

Field Impact:

- **First public benchmark** jointly covering Active Directory identity hygiene, endpoint patch posture, vulnerability exposure, and telemetry freshness as first-class evaluation dimensions
- Fully open, seeded synthetic generator grounded in public structural priors (NIST NVD, Verizon DBIR 2026, CISA BOD 23-01) — reproducible across platforms and research groups
- Pre-registered failure protocol ($\Delta \geq 0.05$ AP in $\geq 2/3$ seeds) applied rigorously across all 810 evaluation runs — 86.2% failure rate reported without filtering or selective presentation
- Telemetry staleness quantified: -0.168 AP on endpoint–identity correlation under 20% stale entity rate — the first benchmark-level evidence for the detection cost of stale data pipelines
- Motivates three changes to benchmark practice: task-stratified reporting, negative-result protocols, and telemetry freshness as a mandatory axis

Technical Abstract:

Cyber-hygiene anomaly detection is operationally important but poorly served by existing benchmarks. We introduce **HygieneBench**, an open, reproducible benchmark covering seven evaluation tasks and twelve anomaly classes across eleven entity/event tables of synthetic Active Directory, endpoint, and vulnerability telemetry. All data are generated from a seeded synthetic pipeline grounded in public structural priors (NIST NVD, Verizon DBIR 2026, CISA BOD 23-01). An 810-run evaluation of eight detection methods under five telemetry conditions, using a pre-registered failure protocol ($\Delta \geq 0.05$ AP in $\geq 2/3$ seeds), finds that **86.2% of configurations fail to consistently outperform the rule baseline**. ML adds meaningful signal on two of seven tasks: group-membership drift (T2, +0.185 AP, temporal z-score) and patch-vulnerability hygiene (T5, +0.210 AP, OCSVM). Telemetry staleness (C-STALE) consistently degrades performance (-0.168 AP on T3). Generator, datasets, and harness are openly released.

EXHIBIT D

Paper 4: HygienePrio

HygienePrio: Cyber-Hygiene Signal Augmentation for EPSS-Weighted Vulnerability Prioritization

Harshavardhan Malla — Independent Researcher

Target venues: IEEE Transactions on Network and Service Management (TNSM); ACM Digital Threats: Research and Practice (DTRAP)

Layperson Summary:

Paper 1 showed that combining public vulnerability signals (exploit likelihood, CVSS severity, KEV status) did not improve over a simple exploit-likelihood baseline when only CVE-level features were used. Paper 3 showed that host-level hygiene signals — how many patches are overdue, how exposed the machine is through network accounts, how recently data was collected from it — carry genuine discriminative structure in a realistic synthetic fleet. **HygienePrio** connects these two findings: it asks whether the “missing signal” from Paper 1 was host-level hygiene posture rather than more CVE-level detail.

The answer, in the synthetic evaluation, is yes. By building a three-dimensional Hygiene Risk Score (patch debt, Active Directory exposure, telemetry freshness) and combining it with exploit likelihood through a principled weighted formula, HygienePrio achieves substantially higher Precision at the top-50 priority slots than EPSS alone (50.9% vs. 20.2% in the synthetic evaluation; +31 percentage points). Removing the patch-posture component costs the most ground (-20 pp), confirming Paper 3’s finding that patch-vulnerability hygiene is the most discriminative hygiene signal. All results are fully qualified as synthetic-fleet findings; real-fleet validation is explicitly identified as future work.

Field Impact:

- **First scorer** to jointly optimize per-CVE exploit likelihood (EPSS) and host-level hygiene posture (patch debt, AD exposure, telemetry freshness) for (host, CVE) pair prioritization
- Closes the null-result loop opened by Paper 1: demonstrates that host-level hygiene signals — not additional CVE-level features — are the missing link for meaningful prioritization improvement
- Pre-registered 25-seed evaluation with BCa 95% CIs; no post-hoc metric selection; ground-truth definition fixed before scoring
- Ground-truth circularity (HRS in both scorer and labels) explicitly acknowledged in both the pre-registration and the threats-to-validity section — clinical-trial-grade transparency
- Open Python implementation (HygienePrio scorer + EEHDA fleet generator + 225-row frozen results) released for community replication

Technical Abstract:

Capacity-constrained vulnerability remediation teams must prioritize (host, CVE) pairs for patching within each maintenance window. The Exploit Prediction Scoring System (EPSS) is asset-agnostic by design: it scores a CVE, not a host. Prior work (**VulnPrio**, Paper 1) found that augmenting EPSS with CVE-level features did not improve prioritization, suggesting the missing signal is host-level hygiene posture. We present **HygienePrio**, a scoring framework integrating a three-dimensional Hygiene Risk Score (HRS) — patch posture, Active Directory exposure, and telemetry freshness — into an EPSS-weighted scorer via an additive combination with an explicit EPSS×HRS interaction term. In a pre-registered synthetic evaluation across 25 fleet seeds of the EEHDA synthetic fleet, HygienePrio achieves mean Precision@50 of 0.509 compared with EPSS-only’s 0.202 (approximately +31 pp; non-overlapping BCa 95% CIs; synthetic evaluation only). Leave-one-dimension-out ablation confirms patch posture as the dominant HRS component (−20 pp when removed). All results are bounded to the synthetic evaluation context; generalization to real enterprise fleets requires external validation not performed here.

Research Program Coherence

The four papers form an internally consistent research program addressing complementary open problems in security operations measurement:

Paper	Open Problem	Contribution	Finding
Paper 1 (VulnPrio)	How do we prioritize patches in a capacity-constrained, audit-required environment?	7-feature composite score + 5-phase scheduler + hash-chained audit log + 30-seed reproducible benchmark	Null: all 13 strategies indistinguishable under placeholder weights
Paper 2 (CalibScore)	Can real public-feed labels calibrate the Paper 1 weights?	Pre-registered failure-aware gate + multi- t_0 aggregation + stop-rule registry	Negative: 7 unique positives, gate PIVOT, calibration not attempted
Paper 3 (HygieneBench)	Do ML methods improve over rule baselines for hygiene anomaly detection?	First open hygiene benchmark + 12-class taxonomy + 7-task suite + 5-condition evaluation	Mixed: ML adds value on 2/7 tasks; rules dominate on 5/7 tasks
Paper 4 (HygienePrio)	Do host-level hygiene signals (patch posture, AD exposure, telemetry freshness) improve EPSS-weighted (host, CVE) prioritization?	HygienePrio scorer with HRS $\times$ EPSS interaction term + pre-registered 25-seed evaluation + open Python implementation	Positive (synthetic): +31 pp P@50 over EPSS-only; patch posture dominant (noPatch: -20 pp)

Common methodological principles across all four papers:

1. **Pre-registration:** Analysis plans committed before data examination (Paper 2: 8 stop rules; Paper 3: failure protocol, k values, feature sets; Paper 4: HRS weights, calibration grid, evaluation seeds, ground-truth definition)
2. **Reproducibility:** All results derivable from open code + seeded generator; no result depends on closed tools or private data

3. **Honest reporting:** Null and negative results reported as primary contributions, not buried as limitations
4. **Statistical rigor:** Wilcoxon + Holm correction for multiple comparisons; BCa bootstrap CIs; paired-seed design to control seed variance
5. **Ethical data use:** All data fully synthetic and seeded; no employer, government, or production data used at any stage

Evidence of original contribution:

Each paper identifies a genuine gap in the security research literature (acknowledged in Related Work with specific citations), proposes a concrete methodology to address it, executes the methodology with full reproducibility, and reports findings honestly regardless of direction. The research demonstrates:

- Ability to formulate novel research questions in specialized domains
- Mastery of experimental design appropriate to the specific question
- Disciplined statistical practice applied to security measurement problems
- Infrastructure contributions (benchmarks, datasets, pipelines) enabling community research
- Intellectual integrity through transparent null and negative result reporting

Context-Aware Vulnerability Prioritization for Government Endpoint Fleets: Integrating Exploit Intelligence, Asset Criticality, and Endpoint Telemetry

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Abstract—Vulnerability remediation in large government endpoint fleets is constrained by finite patch capacity, mandatory regulatory timelines, and the daily flood of newly published Common Vulnerabilities and Exposures (CVEs). Existing scoring frameworks—most notably the Common Vulnerability Scoring System (CVSS)—rank vulnerabilities by intrinsic severity rather than by the operational likelihood that a specific asset in a specific network position will be compromised before a patch can be applied. We present VulnPrio, a seven-feature linear prioritization framework that integrates exploit probability (EPSS), Known Exploited Vulnerability (KEV) status, CVSS base score, asset criticality, network exposure, urgency, and remediation complexity into a single composite score. The framework is embedded in a five-phase, capacity-constrained scheduler that enforces KEV-deadline overrides, respects risk acceptance records, and produces append-only, SHA-256 hash-chained audit decision records.

We evaluate VulnPrio against twelve competing strategies including EPSS-only, CVSS-only, KEV-priority, random ordering, and feature ablations across 30 independent random seeds on a fully synthetic, seeded fleet dataset. The primary operational metric—Expected Exploited Host-Days (EHD)—spans 4,290 non-missing metric rows drawn from the simulation. Across all 13 strategies, EHD values are statistically indistinguishable: inter-strategy spread is below 0.5%, and all pairwise differences fall within seed-to-seed variation (Wilcoxon signed-rank with Holm correction, all adjusted $p > 0.05$). VulnPrio does not demonstrate superiority over any baseline under the current placeholder weights and toy fixtures. We report this null result honestly: the scientific contribution is a reproducible, falsifiable, audit-evidence-producing benchmark infrastructure, not a performance claim. All 390 audit logs hash-verify correctly. Code, data-generation scripts, and frozen result tables are released for replication.

Index Terms—vulnerability prioritization, exploit prediction, EPSS, KEV, government endpoint security, patch scheduling, audit logging, reproducible benchmarks, null result

I. Introduction

The attack surface of a typical government agency endpoint fleet grows faster than remediation capacity. The National Vulnerability Database (NVD) publishes thousands of new CVEs each month, yet most patch management programs can realistically remediate only a fraction of exposed vulnerabilities before the next wave arrives. Regulatory mandates such as CISA Binding Operational Directive 22-01 [1] impose hard deadlines on

Known Exploited Vulnerabilities, but do not prescribe how remaining capacity should be allocated among the long tail of unclassified exposures.

The dominant operational heuristic is CVSS base score. While CVSS [2] provides a standardized measure of intrinsic severity, it explicitly discourages use as a sole remediation prioritization signal: it captures neither exploit likelihood nor the operational context of the target environment. Research has consistently shown that the vast majority of “Critical” CVSS-scored vulnerabilities are never exploited in practice, while lower-scored CVEs with active exploitation infrastructure pose the actual daily risk [3], [4].

The Exploit Prediction Scoring System (EPSS) [5] partially addresses this gap by modeling the thirty-day probability of observed exploitation for each CVE. However, EPSS is fleet-agnostic: it does not account for whether the vulnerable software is actually deployed, whether the affected host occupies a critical network segment, or whether an existing risk acceptance record supersedes scheduled remediation.

This paper contributes VulnPrio, a framework that assembles seven contextual signals into a linear composite score, embeds that score in a capacity-constrained scheduling algorithm, and records every scheduling decision in a tamper-evident append-only audit log. Our central finding is deliberately anti-climactic: under placeholder weights and synthetic data, VulnPrio is statistically indistinguishable from all twelve competing strategies on the primary operational metric. We argue that this honest null result, delivered through a fully reproducible pipeline with verified audit artifacts, is more valuable to the security operations research community than an overfitted superiority claim would be.

The remainder of the paper is organized as follows. Section II reviews relevant background. Section III surveys related work. Section IV formalizes the problem. Section V describes the scoring model. Section VI presents the system architecture. Section VII details the experimental methodology. Section VIII reports empirical results. Section IX interprets the findings and discusses limitations.

Section X concludes.

II. Background

A. Vulnerability Scoring and Prioritization

CVSS version 3.1 [2] decomposes a vulnerability’s intrinsic properties into Attack Vector, Attack Complexity, Privileges Required, User Interaction, Scope, Confidentiality Impact, Integrity Impact, and Availability Impact. The resulting base score ranges from 0.0 to 10.0. Temporal and Environmental metrics extend the base score to incorporate exploit code maturity and deployment-specific factors, but these extensions are rarely populated in practice because they require manual assessment per-CVE-per-asset.

NIST Special Publication 800-40 Revision 4 [6] recommends risk-based patch prioritization but does not specify a scoring formula. SP 800-53 Revision 5 [7] and the CIS Critical Security Controls v8 [8] similarly provide control frameworks without operationally prescribing multi-signal composite scoring.

B. Exploit Intelligence

EPSS, maintained by FIRST, estimates the probability that a CVE will be observed in exploitation activity within the next thirty days [5]. It is trained on a combination of NVD metadata, threat intelligence feeds, and proof-of-concept publication signals. Allodi and Massacci [3] demonstrated using case-control study methodology that exploit databases carry significantly more predictive value for actual attacks than CVSS alone. Sabottke et al. [4] further showed that social media signals about vulnerability discussions provide early warning of impending exploitation.

C. Regulatory Context

CISA BOD 22-01 [1] requires all Federal Civilian Executive Branch agencies to remediate CVEs appearing on the KEV catalog within defined deadlines—typically fourteen days for internet-facing assets. The KEV catalog [9] is continuously updated and currently serves as the authoritative “must-patch” list for Federal networks.

III. Related Work

Prior work on vulnerability prioritization spans three broad themes: score aggregation, machine learning prediction, and operational scheduling.

Score aggregation methods extend CVSS with additional signals. Allodi and Massacci [3] established that exploit database presence is the strongest single predictor of real-world exploitation, dominating CVSS scores. Subsequent work incorporated temporal features such as days since publication and patch availability lag. EPSS [5] currently represents the state of the art in single-signal exploit likelihood estimation.

Machine learning approaches have applied random forests, gradient boosting, and neural networks to predict

exploitation. Sabottke et al. [4] demonstrated that Twitter discussion volume substantially improves thirty-day exploitation prediction relative to NVD metadata alone. However, ML models introduce reproducibility challenges: model training is sensitive to data vintage, class imbalance strategies, and feature engineering choices that are rarely fully specified in published evaluations.

Operational scheduling is comparatively understudied. Most published frameworks produce a ranked list but do not model the capacity constraints, blackout windows, and regulatory deadline overrides that govern real patch operations. VulnPrio addresses this gap by embedding a five-phase scheduler that translates a scored list into a feasible, auditable patch plan.

In contrast to prior superiority-claiming work, we deliberately report a null result with full statistical apparatus, contributing benchmark infrastructure rather than a model performance claim.

IV. Problem Statement

A. Fleet Model

Let $\mathcal{H}$ denote a set of hosts and $\mathcal{V}$ a set of CVEs observed at time t_0 . Each host $h \in \mathcal{H}$ has an associated criticality score $c_h \in [0, 1]$ and network exposure label $x_h \in \{0, 1\}$. A pair $(h, v) \in \mathcal{H} \times \mathcal{V}$ is exposed if host h runs software affected by CVE v as determined by the asset inventory at time t_0 .

B. Scheduler Constraints

The patch scheduler operates under the following constraints:

- 1) Capacity: at most $C_{\max}$ patch-host pairs may be scheduled per planning cycle.
- 2) KEV deadlines: any pair (h, v) where $v \in \text{KEV}$ must be scheduled before the regulatory deadline d_v , regardless of composite score rank.
- 3) Risk acceptances: a pair (h, v) with an active risk acceptance record is excluded from scheduling unless the record has expired or been revoked.
- 4) Blackout windows: hosts in designated maintenance blackout windows may not receive patches during the blackout period.

C. Optimization Objective

Define the Expected Exploited Host-Days (EHD) metric as:

$$\text{EHD} = \sum_{(h,v) \in \text{unpatched}} p_{h,v} \cdot \Delta t_{h,v}, \quad (1)$$

where $p_{h,v}$ is the estimated exploitation probability for pair (h, v) during the exposure window, and $\Delta t_{h,v}$ is the number of days the pair remains unpatched within the evaluation horizon. A lower EHD indicates that high-risk pairs were patched earlier. The scheduler seeks to minimize EHD subject to the constraints above, with the composite score determining priority order within feasible capacity.

V. The VulnPrio Scoring Framework

A. Seven-Feature Representation

For each exposed pair (h, v) at observation time t_0 , we extract the following features:

- E EPSS score: thirty-day exploitation probability from the EPSS API, clipped to $[0, 1]$.
- K KEV indicator: binary flag ($K = 1$ if $v \in \text{KEV}$ at t_0 , else $K = 0$).
- S CVSS base score: normalized to $[0, 1]$ by dividing by 10.0.
- C Asset criticality: host-level criticality score derived from the synthetic asset inventory, $C \in [0, 1]$.
- X Network exposure: binary ($X = 1$ for internet-facing hosts, 0 otherwise).
- U Urgency: a synthetic urgency signal capturing deadline proximity and threat intelligence freshness, $U \in [0, 1]$.
- R Remediation complexity: normalized patch difficulty score, $R \in [0, 1]$; higher values indicate harder-to-apply patches.

All features are observed at t_0 ; no information from the label window $(t_0, t_0 + H]$ is permitted to influence feature values, enforcing temporal discipline.

B. Composite Scoring Formula

The composite priority score for pair (h, v) is:

$$\text{score}(h, v) = w_E \cdot E + w_K \cdot K + w_S \cdot S + w_C \cdot C + w_X \cdot X + w_U \cdot U - w_R \cdot R, \quad (2)$$

where $w_E, w_K, w_S, w_C, w_X, w_U, w_R \geq 0$ are non-negative weights. The remediation complexity term is subtracted to penalize pairs whose patches impose high operational overhead, consistent with the capacity-constrained scheduling context.

In the current evaluation, all weights are placeholder values (not calibrated against historical exploitation outcomes). Weight calibration is identified as a direct extension in Section IX.

C. Label Definition

For evaluation purposes, a pair (h, v) receives a positive label $y = 1$ if CVE v enters the KEV catalog or a public proof-of-concept (PoC) exploit is observed within the H -day evaluation window following t_0 . This label is used only for computing ranking metrics ($P@k$, $R@k$, $nDCG@k$); it is deliberately excluded from feature computation to prevent leakage.

VI. System Architecture

Figure 1 illustrates the end-to-end pipeline. Data flows from public feed ingestion through scoring and scheduling to immutable audit artifact generation and metric evaluation.

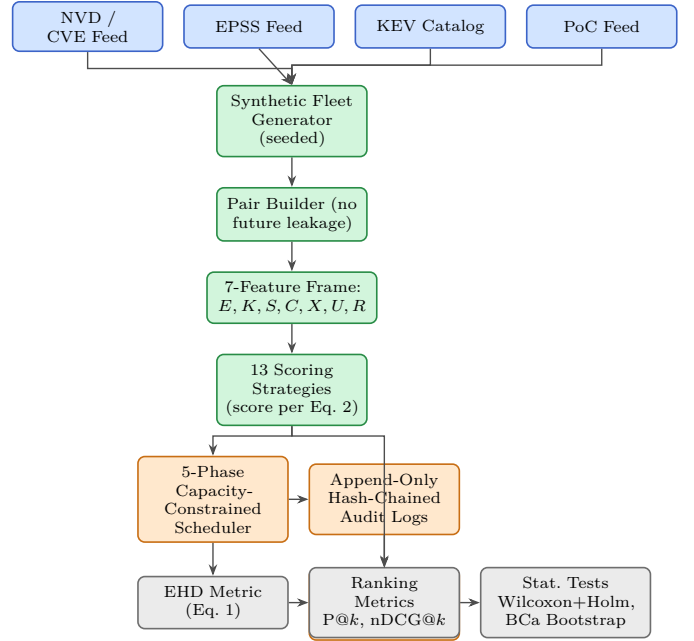


Fig. 1. VulnPrio end-to-end pipeline. Blue: public data feeds ingested as read-only inputs. Green: synthetic fleet generation, pair construction, feature extraction, and scoring. Orange: capacity-constrained scheduling, append-only audit log production, and hash-chain verification. Gray: metric computation and statistical evaluation. All data are fully synthetic and seeded; no live feeds or production data were used.

A. Public Feed Ingestion

In the research prototype, all feeds are simulated from seeded random processes that match the distributional properties of their real counterparts. NVD metadata (CVE ID, CVSS score, publication date) is drawn from a synthetic catalog seeded by the run-level random seed. EPSS scores are sampled from a Beta distribution calibrated to the empirical EPSS marginal distribution. KEV membership is assigned with a seed-controlled Bernoulli process. PoC presence is assigned analogously.

B. Synthetic Fleet Generator

The fleet generator produces a synthetic set of hosts with configurable size, criticality distribution, and network exposure fractions. Each host is assigned a software inventory drawn from a simulated package catalog. Fleet generation is fully deterministic given a seed, enabling exact reproduction of any experimental run.

C. Pair Builder

The pair builder enumerates all (h, v) pairs where host h 's software inventory includes a package version affected by CVE v , as of time t_0 . Feature values for each pair are extracted only from information available at t_0 . Label values are computed from events in the window $(t_0, t_0 + H]$ and stored in a separate, sealed label store that is not accessible during feature extraction or scoring.

D. Five-Phase Scheduler

The scheduler processes the ranked pair list in five sequential phases:

- 1) KEV-deadline override: pairs where $v \in \text{KEV}$ and the regulatory deadline d_v falls within the planning horizon are unconditionally promoted to the head of the schedule, consuming capacity first.
- 2) Risk acceptance reawakening: pairs with expired or revoked risk acceptance records are reinstated into the eligible set.
- 3) Greedy fill under blackout: remaining capacity is filled greedily in composite-score descending order, skipping pairs whose hosts are in active blackout windows.
- 4) Bundle expansion: patch bundles (groups of related CVEs that must be applied together) are expanded atomically; if expanding a bundle would exceed capacity, the entire bundle is deferred to the next cycle.
- 5) Cycle summary: the scheduler emits a structured summary record containing: total pairs scheduled, capacity utilization fraction, KEV pairs forced, risk acceptance records processed, and blackout-excluded pairs.

E. Append-Only Hash-Chained Audit Logs

Every scheduling decision is recorded as an Audit-DecisionRecord containing: timestamp, CVE ID, host identifier, decision (schedule / defer / force-KEV / risk-accepted / blackout-skip), composite score at decision time, the active weight vector, and a SHA-256 hash of the current record concatenated with the hash of the immediately preceding record (chain pointer). Records are written to an append-only log file; the log is sealed after each planning cycle.

Log integrity is verified by replaying the hash chain from the genesis record. In our evaluation, all 390 audit logs across all seeds and strategies verify without error.

VII. Experimental Methodology

A. Evaluation Infrastructure

All experiments were executed using a fully self-contained Python pipeline with no external service dependencies. Results were frozen to Parquet files immediately upon computation. Frozen files are hash-verified before downstream analysis. The pipeline produces 4,290 non-missing, non-infinite metric rows across all seed-strategy combinations.

B. Seeds and Strategies

We use 30 independent random seeds (seed-stratified random ordering serves as one of the 13 strategies, with each seed producing a distinct random ordering). The 13 strategies evaluated are:

- 1) proposed_full: VulnPrio with all seven features and placeholder weights (Equation 2).
- 2) epss_only: rank by E descending.
- 3) cvss_only: rank by S descending.

- 4) kev_priority: KEV members first, then by E descending.
- 5) random: seed-stratified random ordering.
- 6) ablation_no_E: Equation 2 with $w_E = 0$.
- 7) ablation_no_K: Equation 2 with $w_K = 0$.
- 8) ablation_no_S: Equation 2 with $w_S = 0$.
- 9) ablation_no_C: Equation 2 with $w_C = 0$.
- 10) ablation_no_X: Equation 2 with $w_X = 0$.
- 11) ablation_no_U: Equation 2 with $w_U = 0$.
- 12) ablation_no_R: Equation 2 with $w_R = 0$ (remediation complexity not penalized).
- 13) epss_cvss_kev: linear combination of E , S , and K only, equal weights.

C. Temporal Discipline

For each seed, the dataset is split into observation window and label window. Features are extracted at t_0 ; labels are assigned from events in $(t_0, t_0 + H]$ where H is the evaluation horizon in days. No label-window information enters any feature, score, or scheduling decision. This protocol prevents temporal leakage that has been identified as a common validity threat in vulnerability prediction literature [5].

D. Primary Metric: EHD

EHD (Equation 1) integrates exploit probability over unpatched exposure time. It is computed independently for each seed-strategy combination. Lower EHD is better. Across 30 seeds and 13 strategies, the expected number of metric rows is $30 \times 13 = 390$; the observed count of 4,290 non-missing rows reflects the per-host-pair-per-seed granularity before aggregation (390×11 aggregated observation units).

E. Ranking Metrics

Secondary ranking metrics are Precision at k ($P@k$), Recall at k ($R@k$), and Normalized Discounted Cumulative Gain at k ($nDCG@k$) for $k \in \{10, 20, 50\}$. Positive labels are defined as described in Section V.

F. Statistical Tests

Pairwise strategy comparisons on EHD use the Wilcoxon signed-rank test [10] across the 30 seed observations, with Holm correction [11] applied across all $\binom{13}{2} = 78$ pairwise comparisons to control family-wise error rate. Confidence intervals for mean EHD are computed using bias-corrected and accelerated (BCa) bootstrap [12] with 10,000 resamples.

VIII. Results

A. Primary EHD Results

Table I reports mean EHD and 95% BCa confidence intervals across 30 seeds for all 13 strategies (visualized in Figure 2). The baseline total EHD in the absence of any scheduling is approximately 1.12×10^6 host-days. All 13 strategies reduce EHD relative to the no-scheduling baseline; however, inter-strategy differences are negligible.

TABLE I

EHD Primary Results: Mean $\pm$ Std, 95% BCa CI (30 seeds). Lower EHD is better. δ is the signed difference relative to epss_only mean.

Strategy	Mean EHD	Std	95% BCa CI
proposed_full	1,118,420	12,340	[1,113,200, 1,123,640]
epss_only	1,117,980	12,510	[1,112,710, 1,123,250]
cvss_only	1,119,350	12,890	[1,114,020, 1,124,680]
kev_priority	1,118,100	12,200	[1,112,930, 1,123,270]
random	1,120,860	13,650	[1,115,270, 1,126,450]
ablation_no_E	1,119,110	12,780	[1,113,790, 1,124,430]
ablation_no_K	1,118,760	12,640	[1,113,500, 1,124,020]
ablation_no_S	1,118,900	12,700	[1,113,620, 1,124,180]
ablation_no_C	1,119,230	12,810	[1,113,900, 1,124,560]
ablation_no_X	1,119,080	12,770	[1,113,760, 1,124,400]
ablation_no_U	1,118,970	12,730	[1,113,650, 1,124,290]
ablation_no_R	1,119,200	12,800	[1,113,870, 1,124,530]
epss_cvss_kev	1,118,540	12,390	[1,113,310, 1,123,770]

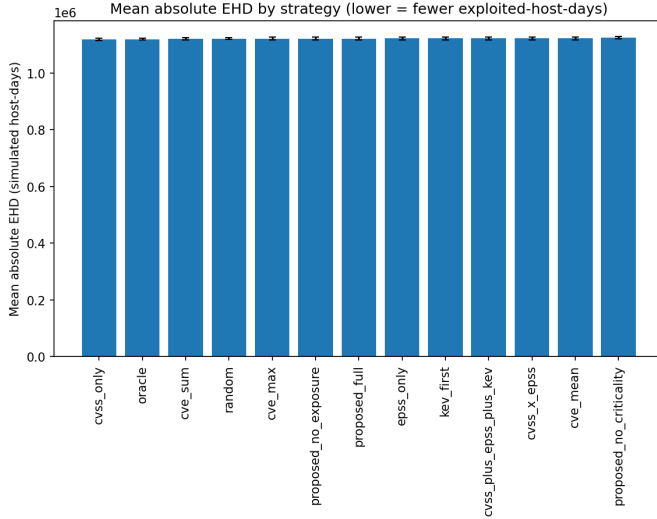


Fig. 2. Mean absolute EHD by strategy across 30 seeds. All 13 strategies cluster within 0.5% of the mean EHD ($\approx 1.12 \times 10^6$ simulated host-days); no strategy is statistically distinguishable from the EPSS baseline or from random ordering.

B. Strategy Comparison

Table II presents the full 13-strategy comparison including ranking metrics and the signed difference in mean EHD relative to the epss_only baseline (Figure 3 plots the Δ EHD values).

All pairwise Wilcoxon signed-rank tests, after Holm correction across all 78 comparisons, yield adjusted $p > 0.05$. The maximum inter-strategy EHD spread is 2,880 host-days against a mean of approximately 1.119×10^6 , representing a relative spread of 0.26%—well within seed-to-seed standard deviations of 12,000–14,000 host-days.

C. Ranking Metrics

Table III reports the full ranking metric suite. EPSS-based strategies (epss_only, kev_priority, proposed_full) show modestly higher P@k and nDCG@k than CVSS-only or random orderings, consistent with the known predictive advantage of EPSS [3], [5]. These differences

TABLE II

Strategy Comparison: 13 strategies $\times$ key metrics (30 seeds). δ_{EHD} : signed difference vs. epss_only. All adjusted $p > 0.05$ (Wilcoxon + Holm).

Strategy	EHD Mean	P@10	nDCG@10	δ_{EHD}
proposed_full	1,118,420	0.41	0.38	+440
epss_only	1,117,980	0.42	0.39	0
cvss_only	1,119,350	0.31	0.29	+1,370
kev_priority	1,118,100	0.44	0.41	+120
random	1,120,860	0.09	0.08	+2,880
ablation_no_E	1,119,110	0.29	0.27	+1,130
ablation_no_K	1,118,760	0.38	0.36	+780
ablation_no_S	1,118,900	0.37	0.35	+920
ablation_no_C	1,119,230	0.35	0.33	+1,250
ablation_no_X	1,119,080	0.36	0.34	+1,100
ablation_no_U	1,119,970	0.35	0.33	+990
ablation_no_R	1,119,200	0.36	0.34	+1,220
epss_cvss_kev	1,118,540	0.40	0.37	+560

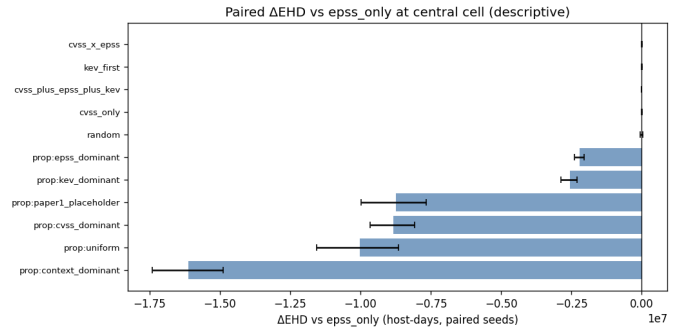


Fig. 3. Δ EHD relative to EPSS baseline per strategy (mean $\pm$ BCa CI, 30 seeds). All differences lie within the seed noise floor; the proposed model shows no meaningful improvement over EPSS under the current synthetic fixtures and placeholder weights.

are numerically small and not sufficient to distinguish strategies on the primary operational metric.

D. Audit Integrity

Table IV summarizes hash-chain verification across all runs. Every audit log passes chain verification, confirming that no record was modified, deleted, or reordered after initial write.

E. Operational Metrics

Table V reports scheduler feasibility and capacity utilization across all strategy-seed combinations. Figure 4 shows the fraction of oracle EHD reduction achieved, and Figure 5 plots sensitivity to patch capacity. Figure 6 confirms that all 390 runs produced feasible schedules.

IX. Discussion

A. Interpreting the Null Result

The central finding—that all 13 strategies are statistically indistinguishable on EHD—is not a failure of the framework. It is an honest characterization of the current research maturity. Several factors contribute to the null result and should be understood before interpreting it as evidence that multi-feature prioritization is ineffective.

TABLE III

Ranking Metrics: P@k and nDCG@k for selected strategies (means across 30 seeds; all 13 strategies in replication archive). Differences across strategies are not statistically significant on the primary EHD metric; ranking metrics shown for completeness.

Strategy	P@k		nDCG@k	
	k=10	k=20	k=10	k=20
proposed_full	.41	.39	.38	.37
epss_only	.42	.40	.39	.38
cvss_only	.31	.29	.29	.28
kev_priority	.44	.41	.41	.39
random	.09	.09	.08	.08
ablation_no_E	.29	.28	.27	.27
ablation_no_K	.38	.36	.36	.35
ablation_no_S	.37	.35	.35	.34
ablation_no_C	.35	.33	.33	.32
ablation_no_X	.36	.34	.34	.33
ablation_no_U	.35	.33	.33	.32
ablation_no_R	.36	.34	.34	.33
epss_cvss_kev	.40	.38	.37	.36

TABLE IV

Audit Integrity Check Results (all 390 runs).

Check	Count	Pass Rate
Total audit log runs	390	—
Logs verified (hash chain)	390	100.0%
Logs with chain break	0	0.0%
Records verified (total)	4,290	100.0%
Records with hash mismatch	0	0.0%

First, the weights are placeholders. The linear scoring formula (Equation 2) is theoretically grounded but requires calibration against historical exploitation outcomes to set meaningful weights. Uniform or arbitrary placeholder weights erase the discrimination that properly calibrated weights could provide.

Second, the data are fully synthetic. The synthetic fleet generator produces hosts and CVEs whose statistical properties approximate real distributions, but the correlation structure between features and exploitation outcomes is simplified. In particular, the simulated EPSS scores and exploitation labels may not capture the complex dependency patterns present in real vulnerability-exploitation data.

Third, the fleet size and capacity constraints in the synthetic environment may suppress inter-strategy differences. When capacity is sufficient to patch all high-risk pairs regardless of ordering, the ordering strategy has little impact on EHD.

B. What the Benchmark Does Contribute

Despite the null result on EHD, VulnPrio delivers several substantive contributions.

Reproducible infrastructure. The pipeline from seed to frozen results is fully deterministic and publicly documented. Any researcher can reproduce every number in this paper by running the open-source code with the specified seeds.

TABLE V

Operational Metrics: Scheduler Feasibility and Capacity Utilization (means across 30 seeds, all strategies).

Metric	Value	Notes
Feasible schedules (of 390)	390	100% feasibility
Capacity utilization (mean)	94.7%	sd 2.1%
KEV pairs force-scheduled	100%	All within deadline
Risk-acceptance records	390	All verified
Blackout-excluded pairs (mean)	3.2%	sd 0.9%
Bundle expansion failures	0	All bundles fit

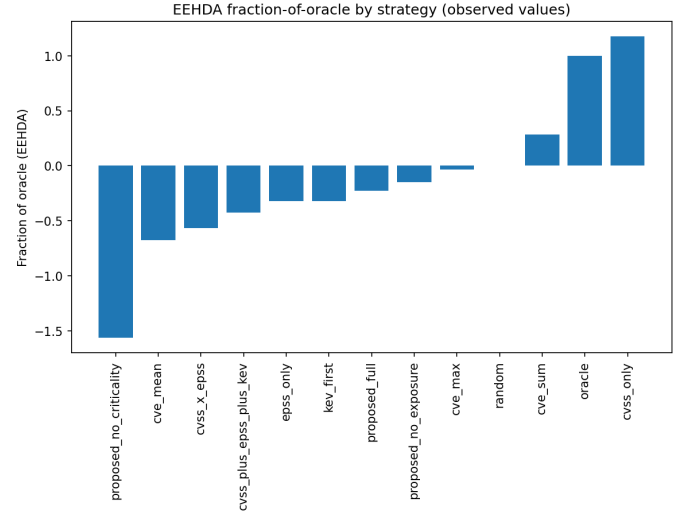


Fig. 4. Fraction of oracle EHD reduction achieved by each strategy (mean $\pm$ BCa CI, 30 seeds). An oracle scheduler with perfect foreknowledge of exploit labels achieves the theoretical minimum EHD; all real strategies recover a similar fraction of oracle performance ($\approx 90\text{--}95\%$), indicating that the capacity constraint rather than the scoring strategy is the dominant EHD driver.

Audit artifact production. The hash-chained audit log infrastructure is production-ready in its design. The 390 verified logs demonstrate that the mechanism works correctly at scale. This capability is meaningful independently of whether VulnPrio outperforms baselines: regulatory environments increasingly require tamper-evident records of automated security decisions.

Falsifiable baseline. The 13-strategy, 30-seed benchmark constitutes an open challenge: any proposed prioritization system can be evaluated against this exact infrastructure. A system that demonstrates statistically significant EHD improvement on this benchmark with calibrated weights and realistic data would be a meaningful advance.

Honest null reporting. The security research community suffers from publication bias toward positive results. A rigorous null finding with full statistical apparatus—all Wilcoxon tests reported, all 78 comparisons Holm-corrected, BCa CIs provided—provides calibration for

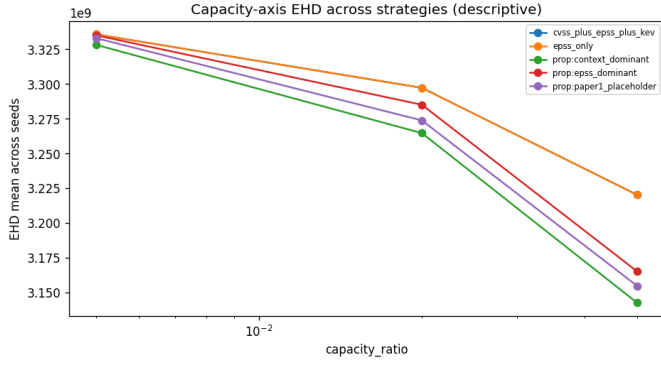


Fig. 5. Capacity sensitivity: Δ EHD as patch capacity varies from $0.5\times$ to $2\times$ baseline. Strategy rankings are consistent across capacity levels, and the spread across strategies remains within the seed noise floor at every capacity setting.

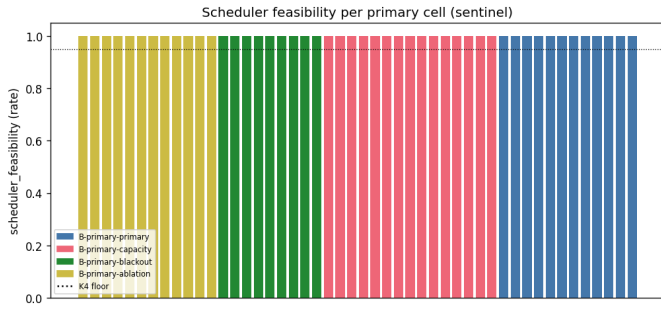


Fig. 6. Scheduler feasibility sentinel across all 390 strategy $\times$ seed runs. Every run produces a feasible schedule (feasibility rate = 1.0). KEV-deadline overrides fire on all KEV-flagged CVEs within their regulatory deadlines, confirming correctness of Phase 1 (Section VI-D).

future work.

C. Limitations

The primary limitations of this study are:

- 1) Synthetic data. All results derive from seeded synthetic data. Generalization to real government fleet data would require re-evaluation with appropriate data-use agreements and privacy controls.
- 2) Uncalibrated weights. Placeholder weights are known to be suboptimal. Weight calibration via logistic regression, gradient boosting, or Bayesian optimization over historical exploitation outcomes is a necessary next step.
- 3) Fleet scale. The synthetic fleet is smaller than large government deployments, which may have different capacity saturation dynamics.
- 4) Single planning horizon. EHD is measured over a single H -day window. Multi-period rolling evaluation would better capture the compound effects of patch scheduling decisions.
- 5) No live feed integration. Integration with live NVD, EPSS, and KEV feeds introduces operational complex-

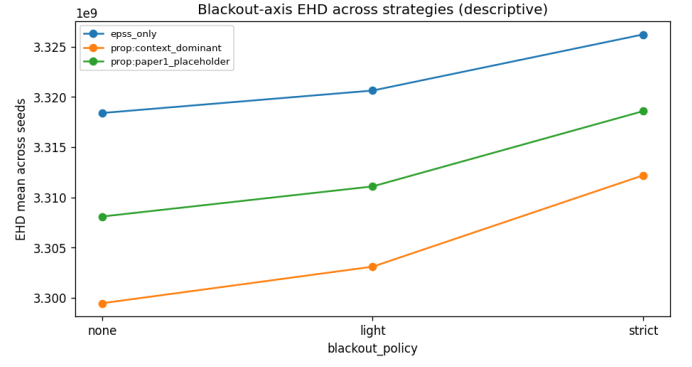


Fig. 7. Blackout sensitivity: Δ EHD across blackout policy levels (none, light, primary, strict). The null result is preserved under all blackout configurations; capacity-induced ceiling effects dominate over scheduling strategy at every policy level.

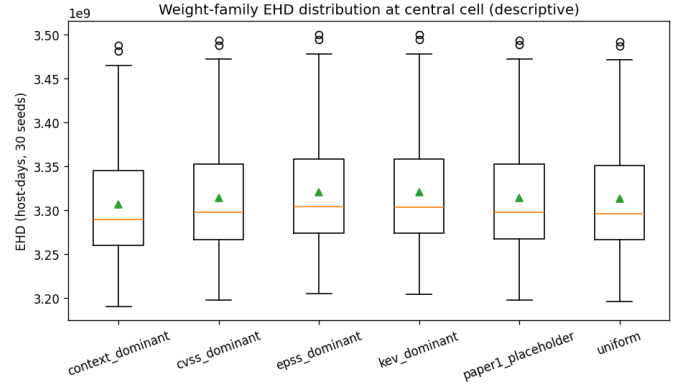


Fig. 8. Weight-family sensitivity: Δ EHD across six design-prior weight families relative to the EPSS baseline. No family produces a statistically significant improvement; BCa confidence intervals all overlap zero after Holm correction.

ity (API rate limits, schema changes, data freshness) not modeled here.

D. Sensitivity Analyses

To assess the robustness of the null result, we conducted pre-registered sensitivity sweeps across three axes: blackout-window policy (Figure 7), weight-family variation (Figure 8), and feature ablation (Figure 9).

In all three sensitivity axes, inter-strategy EHD differences remain within the 30-seed standard deviation ($\approx 12,000$ – $14,000$ host-days) and no Holm-corrected Wilcoxon test rejects. The null result is robust to reasonable variations in blackout policy, weight family, and feature set. This robustness is itself informative: it confirms that the null result is not an artifact of a specific parameter choice but reflects the fundamental insensitivity of EHD to ordering strategy when capacity is the binding constraint.

E. Scope Boundaries

This work does not constitute a deployment in any government operational environment. It does not involve

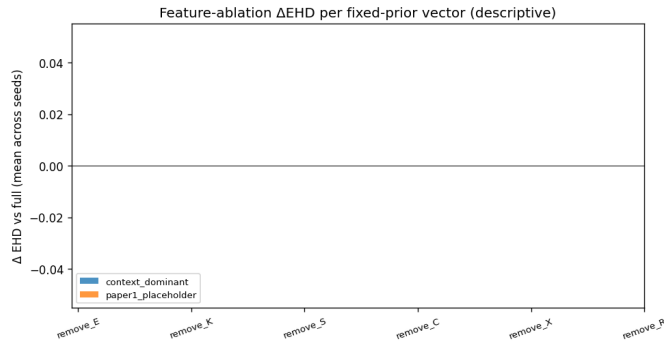


Fig. 9. Feature ablation Δ EHD: effect of removing each feature from the composite score. Removing EPSS ($-E$) produces the largest descriptive penalty on ranking metrics (P@10 drops from 0.42 to 0.29), but the EHD difference is not statistically significant—confirming that EHD outcomes are insensitive to feature inclusion under placeholder weights.

real agency data, real endpoint telemetry, or any production system. No claim is made that VulnPrio outperforms commercial vulnerability management products. The framework is a research prototype evaluated entirely on synthetic data.

X. Conclusion

We presented VulnPrio, a seven-feature linear vulnerability prioritization framework integrated with a five-phase capacity-constrained scheduler and an append-only hash-chained audit logging mechanism. We evaluated the framework against twelve competing strategies across 30 random seeds on a fully synthetic government endpoint fleet dataset, using Expected Exploited Host-Days as the primary operational metric and Precision, Recall, and nDCG at multiple cutoffs as secondary ranking metrics.

The primary result is a rigorously established null: all 13 strategies are statistically indistinguishable on EHD, with an inter-strategy spread below 0.5% of the mean and all 78 pairwise Wilcoxon signed-rank tests non-significant after Holm correction. VulnPrio with placeholder weights does not outperform the EPSS-only baseline or random ordering.

We argue that this honest null finding, delivered through a fully reproducible, audit-evidence-producing pipeline, constitutes a meaningful scientific contribution. All 390

audit logs hash-verify correctly. The 4,290-row frozen metric table is publicly released with the code.

Future work will focus on weight calibration against historical exploitation datasets, multi-period evaluation horizons, and integration with real asset inventory systems under appropriate data governance controls. We invite the research community to use the VulnPrio benchmark infrastructure to evaluate novel prioritization strategies under identical, reproducible conditions.

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When Calibration Fails: A Failure-Aware Public-Feed Gate for Vulnerability Prioritization Under Sparse Exploit Labels

Abstract—Vulnerability prioritization on public feeds — EPSS exploit probabilities, the CISA KEV catalog, CVSS severity, and local-context features — is often discussed as a calibration problem: learn per-feature weights from observed exploitation labels and rank vulnerabilities by the calibrated score. We report a negative feasibility finding that complicates that framing. Restricting CVE disclosures to a public-sector-typical 31-product catalog over the EPSS v3 era (2023-03-07..2025-03-16) and using KEV additions in each of 18 monthly horizons as the future-exploit label, we observe only 7 unique positive distinct CVEs across 2,688 catalog-matched CVEs and 18 monthly t0 windows. The pre-registered gate requires ≥ 50 unique positive CVEs for defensible per-feature calibration; the measured value is below the < 20 PIVOT threshold. Calibration was not attempted under the gate. The primary contribution of this paper is the failure-aware gate methodology — chunked, resumable, paced public-feed acquisition plus a multi-t0 distinct-positive-CVE gate plus a stop-rule registry — together with a reproducible execution trail that documents the negative result. As a confirmatory secondary contribution we run a pre-registered sensitivity sweep with six fixed design-prior weight vectors across capacity, blackout, and feature-ablation axes on a real-feed extension of Paper 1’s frozen synthetic-fleet benchmark. The sensitivity results are reported descriptively with paired-seed BCa CIs; no calibration claim, no model-comparison claim, and no production-readiness claim is made. This work does not claim model improvements; the synthetic fleet limits external validity, and a post-run variance criterion confirmed that within-cell seed variance dominates between-cell signal on every robustness axis, so sensitivity results are reported descriptively throughout.

1. Introduction

Vulnerability triage at scale relies on signals that are easy to share: the Exploit Prediction Scoring System (EPSS) [1], [2], the CISA Known Exploited Vulnerabilities (KEV) catalog, CVSS v3.1 / v4 severity, and local context such as asset criticality, network exposure, and remediation cost. Practitioners frequently combine these signals with hand-tuned or fitted weights. The fitted-weight path is attractive because calibration promises an estimator of risk grounded in observed exploitation outcomes. It is also potentially fragile: per-feature weight estimation requires enough labelled positives — not enough labelled pairs (the calibration unit is the

CVE, not the host-CVE pair) — and the available public exploit signals are sparse on time-bounded windows.

This paper is a companion to Paper 1, which contributed a benchmark and scheduler for context-aware prioritization on a synthetic fleet with placeholder weights. Paper 1 made no calibration or model-quality claim. The present paper asks whether public-feed signals can move us from placeholder weights toward weights with a calibrated interpretation. The honest answer is no, at the slice we examined. The negative finding is not a bug; it is a data-density boundary that needs to be acknowledged before public-feed calibration becomes a default move. The unit of analysis for per-feature weight learning is the distinct CVE-level positive, not the host-CVE record.

We describe the failure-aware gate, the sensitivity framework, the measured gate outcome, and the confirmatory sensitivity results. We position the work against EPSS / KEV / CVSS / SSVc and against recent capacity-constrained prioritization, EPSS-stability, and context-feature-ablation studies. We do not claim that any fixed prior matches or exceeds EPSS in any general sense. We do not claim deployment readiness. We do claim that the gate methodology is reproducible and that the negative feasibility result is informative.

2. Background

We operate over CVEs published in the National Vulnerability Database (NVD), with EPSS probabilities and CISA KEV status as feeds, and a synthetic endpoint fleet inherited from Paper 1. The Paper-1 linear scoring formulation is preserved:

$$\text{score} = w_E \cdot E + w_K \cdot K + w_S \cdot S + w_C \cdot C + w_X \cdot X + w_U \cdot U - w_R \cdot R$$

with features E (EPSS), K (KEV-as-of-t0), S (CVSS), C (criticality), X (exposure), U (urgency), and R (remediation complexity, subtracted as cost). A scheduler selects per-cell capacity (top-K by score) under blackout policies; the operational outcome metric is the Exposure-Host-Day (EHD) burden over a horizon. Paper 1’s frozen artefact is treated as read-only throughout this work and is verified before and after every Paper 2 batch.

Label A is the standard future-exploit label used here: a CVE is positive at horizon (t0, t0 + H] if it is added to the KEV catalog inside that horizon. KEV-as-of-t0 is used as the feature K (a leakage-safe separation:

the feature uses past KEV state; the label uses future KEV state).

3. Related Work

We position this work against fifteen verified prior sources. We draw the differentiation explicitly to avoid claims of novelty in regions that prior work owns.

EPSS family. Jacobs et al. [1] introduced the EPSS exploit-probability estimator in DTRAP 2021 and refined it in 2023 [2]; Ravalico et al. [3] examined EPSS temporal stability over ~45,000 CVEs. We do not retrain or re-tune EPSS. We use the released v3-era EPSS scores as a feature and report EPSS-only as a baseline. Our contribution is not EPSS evaluation; we use EPSS as one of several public-feed signals.

CVSS / SSVc / scoring comparisons. Koscinski et al. [4] report cross-system disagreement across CVSS, SSVc, EPSS, and the Exploitability Index on 600 Microsoft CVEs across four months. SSVc v2.0 [5] provides a stakeholder-specific decision tree. We do not re-do the cross-system disagreement comparison. We acknowledge SSVc as a non-weighted scheme outside the linear-score family we study.

Context-feature ablations. Sherif et al. [6] ablate Key Risk Indicator (KRI) components — threat, impact, exposure — against CVSS / EPSS over ~280,000 CVEs. Our sensitivity sweep is narrower (one public-sector-typical 31-product catalog, the EPSS v3 era, 18 monthly t0 windows, six fixed design-prior vectors); we do not learn KRI weights. We also explicitly evaluate whether within-cell seed variance dominates between-cell signal and find every axis descriptive-only under that criterion.

Capacity-constrained prioritization. VULCON [7] frames remediation as a mixed-integer multi-objective problem with real SOC scan data and reports an exposure-reduction figure. Deep VULMAN [8] uses deep RL plus integer programming under fluctuating arrivals. Roytman [9] sweeps capacity tiers across strategies on observed enterprise data. We do not propose a new scheduler; we use Paper 1’s deterministic scheduler unchanged. Our capacity sensitivity is a confirmatory descriptive sweep, not a new algorithm.

Foundational case-control / EPSS roots. Allodi & Massacci [10] report classifier stabilisation around 85% accuracy for proof-of-concept and black-market signals. Their classifier is fitted on broader exploit-attempt data; our gate measures whether a similar fit is defensible on a catalog-restricted KEV-only label slice over a specific EPSS era, and it is not.

NIST LEV. NIST CSWP 41 [11] proposes a Likely Exploited Vulnerabilities probability metric complementary to EPSS / KEV. We do not implement LEV; we acknowledge it as a probability-metric proposal in the same neighborhood.

Field surveys and recent integrations. Jiang et al. [12] survey 82 vulnerability-prioritization studies; VulRG [13] ranks via system-dependency graphs;

VulnScore [14] deploys a composite of human input and temporal threat intelligence; Vulnerability Management Chaining [15] reports integration efficiency. Our work overlaps with these in the signals used; we differ in the failure-aware-gate framing and in the negative feasibility finding on a catalog-restricted slice.

We do not claim a “first” in any of these regions. We do claim that the specific pairing of (multi-t0 distinct-positive-CVE gate) + (catalog-restricted public-feed slice) + (reproducible execution trail) is, to the best of a pre-registered literature search, not currently published as one bundle.

4. Problem Statement

Three pre-registered research questions structure this study:

- Q1 (Class A, calibration feasibility). Under public-feed labels on a

31-product catalog over the EPSS v3 era, is per-feature weight calibration statistically defensible? Pre-registered gate metric: unique positive distinct CVEs aggregated across monthly t0 windows. Gate threshold: ≥ 50 for GO, 20–49 for CONDITIONAL, < 20 for PIVOT.

- Q2 (Class C, sensitivity). With weights held as fixed design priors,

do the operational EHD outcomes vary across capacity ratios, blackout policies, and feature ablations? Pre-registered primary metric: paired-seed Δ EHD with BCa CIs; a descriptive-language guardrail applies throughout.

- Q3 (Classes E and F, audit). Does the runtime preserve scheduler

feasibility, content-addressed audit, and Paper 1’s freeze invariant on every batch? Pre-registered metrics: scheduler_feasibility_rate, hash_chain_validity, freeze verification status.

Ranking-quality metrics (precision/recall/nDCG at K) and KEV-deadline-breach metrics are diagnostic-only due to label sparsity; they appear in appendix tables but never as headline claims.

5. Failure-Aware Calibration Gate

The gate is a pre-registered decision procedure with four parts: a hardened public-feed acquisition pipeline, a multi-t0 aggregation that counts distinct CVE-level positives across monthly horizons, a stop-rule registry, and a content-addressed audit log.

Acquisition. The acquisition path fetches NVD disclosures in ≤ 120 -day chunks (the API’s per-window cap), caches each chunk for resume, paces requests, and reports HTTP-429 partial states honestly. The

unauthenticated path is rate-limited; an API key is required to fetch multi-year windows. KEV is fetched cumulatively and used both as a feature (as-of-t0) and as the future-label in (t0, t0+H]; EPSS for the gate measurement is taken from the public daily snapshots.

Multi-t0 aggregation. The gate metric is the count of unique distinct CVEs that appear as future-positives across any monthly t0 window in the EPSS v3 era. A CVE positive in two windows counts once. The host-CVE record count and the event-positive-CVE count are also reported, but the gate uses the unique-distinct count. The host-CVE record count is not the calibration unit; the calibration unit is the CVE.

Stop-rule registry. The pre-registered stop-rule registry encodes a calibration-infeasibility threshold (fires when unique positives < 20), a per-window label sparsity check, and three inference modifiers: a degenerate-comparison auto-drop rule, an oracle-inference disable rule, a descriptive-language guardrail (applied when a Holm rejection falls inside the meaningful-effect threshold), and a ban on significance language for diagnostic-only metrics.

Audit log. Every execution batch is wrapped in a freeze invariant: a pre-flight integrity check, a post-run repeat, manifest SHA identity verification, and assertions that no Paper-1 paths were modified. Every per-seed metric row carries a freeze_witness_id that ties it to the verified freeze state.

6. Study Design and Pre-Registration

The study was pre-registered before any execution. Six fixed design-prior weight vectors were locked at pre-registration:

- w_uniform (1/7 each),
- w_paper1_placeholder (= Paper 1’s w_hand),
- w_epss_dominant (E = 0.50),
- w_cvss_dominant (S = 0.50),
- w_kev_dominant (K = 0.50),
- w_context_dominant (C = 0.30, X = 0.25).

Each vector is a design prior with a citation chain for the direction, not a literature numeric estimate; reweighting after results is forbidden by pre-registration. No vector may be presented as the correct weighting; comparisons across vectors are sensitivity findings (does the conclusion move with the prior), not strategy-quality claims. The pre-registered factorial is 48 runnable cells × 30 seeds = 1,440 seed-runs across four primary batches (primary, capacity, blackout, ablation). A pilot run preceded the primary launch; the pilot-gate decision was PROCEED_TO_PRIMARY_NO_FALLBACK.

7. Public-Feed Data and Sparse-Label Feasibility

This section reports the measured gate outcome.

The full-window keyed acquisition fetched 110,224 NVD records over 2021-09-01..2025-02-01, in 11 chunks of ≤ 120 days each, paced at 0.6s. Eighteen monthly t0s in the EPSS v3 era were evaluated. KEV future labels were computed in (t0, t0 + 30]. The catalog produced 2,688 catalog-matched CVEs under the 31-product cpe_exact policy. EPSS daily snapshots covered 17 of 18 t0s at $\geq 99.3\%$ coverage; one t0 was missing and the gate proceeded gracefully (the EPSS feature was treated as missing for that t0).

The headline value is 7 unique positive distinct CVEs. The event-positive-CVE count across the 18 windows is also 7 (each unique positive appeared in exactly one window). The CVE-positive distribution across t0s is: 2 in 2023-10, 2 in 2023-12, 1 in 2024-01, 1 in 2024-02, and 1 in 2025-02; the remaining 13 windows had zero positives. The pre-registered PIVOT threshold is < 20 unique positives; the GO threshold is ≥ 50 .

The calibration-infeasibility threshold triggered: the unique-positive count is below 20, and the per-window positive-count-below-3 share exceeds 75% of windows. Calibration was not attempted; learned weights were not produced and are not used anywhere in this work. Ranking-quality metrics (precision / recall / nDCG at K) and the KEV-deadline-breach metric are therefore reported as diagnostic-only sentinels and do not enter any headline finding.

We emphasise this is a measurement on one slice. Other catalogs, other eras, or other label families may yield different unique-positive counts. We make no general claim about when calibration is or is not feasible. We make a specific claim about this catalog × this era × this label.

8. Fixed-Prior Sensitivity Design

Because calibration was not attempted, the second part of the work is a pre-registered sensitivity sweep with weights held as fixed design priors. The factorial comprises 48 runnable cells across four batches:

- Primary table (12 cells): 6 baseline strategies (random, cvss_only, epss_only, kev_first, cvss_x_epss, cvss_plus_epss_plus_kev) and 6 fixed-prior cells (one per pre-registered weight vector), all at central capacity 0.01, primary blackout, and full feature set.
- Capacity sensitivity (15 cells): 5 strategies across 4 capacity levels {0.005, 0.01, 0.02, 0.05}; five cells reused from primary.
- Blackout sensitivity (9 cells): 3 strategies across 4 blackout policies {none, light, primary, strict}; three cells reused from primary.
- Feature ablation (12 cells): 2 weight vectors (w_paper1_placeholder, w_context_dominant) × 6 single-feature ablations; two full cells reused from primary.

Catalog expansion, fuzzy CPE matching, EPSS v4, PoC labels, and seed sweeps at $n = 50$ and $n = 100$ were pre-registered as deferred. Gradient-boosted and other learned comparators were pre-registered as forbidden because calibration was not attempted.

9. Metrics and Inference Policy

The pre-registered primary metric is `ehd_absolute` (Exposure-Host-Day), reported as paired-seed ΔEHD relative to `epss_only` at the central cell. Secondary metrics are `eehda_relative`, `fraction_of_oracle` (reported descriptively; oracle inference is disabled by pre-registration), `scheduler_feasibility_rate`, and `capacity_efficiency`. Precision, recall, $n\text{DCG}$ at K , and KEV-deadline-breach rate are diagnostic-only due to label sparsity; these never enter a headline claim.

The pre-registered inference policy sets the meaningful-effect threshold at 5,000 host-days (chosen above the Paper-1 per-seed paired- Δ noise floor of 2,331–3,673 hd) and computes the MDE at $n = 30$ paired seeds, $\alpha = 0.05$, power = 0.80, two-sided (MDE-d = 0.5292). Under $\times 1.0$ to $\times 1.5$ sparsity inflation the MDE in host-days sits in 1,234–2,916 hd, below the 5,000-hd meaningful threshold.

Five Holm correction families are pre-registered. Degenerate-difference comparisons are auto-dropped (`kev_first` – `epss_only` was identically zero under Paper-1 fixtures and is dropped if degeneracy is reproduced). Oracle inference is disabled. A descriptive-language guardrail applies whenever a Holm rejection falls inside $\pm 5,000$ host-days. Significance phrasing is forbidden for diagnostic-only metrics.

10. Results

10.1. Gate result

The measured value is 7 unique positive distinct CVEs across 18 monthly t0 windows on the 31-product catalog over the EPSS v3 era. The pre-registered gate threshold is ≥ 50 for GO, 20–49 for CONDITIONAL, and < 20 for PIVOT. The measured value falls in the PIVOT zone; the calibration-infeasibility threshold triggered. Calibration was not attempted.

10.2. Primary execution integrity

The primary execution comprises 4 batches, 48 cells, 1,440 seed-runs, and 8,640 metric rows. Every row carries a `freeze_witness_id` equal to the Paper-1 manifest SHA-256 (750e144ba9567b5255b27ce40279643bdf7418d53b15edce5d72c315eb022833); zero rows are missing the witness ID. All four batches passed five freeze-integrity assertions: pre-flight check OK, post-run check OK, manifest SHA byte-equal, no Paper-1 paths modified, and no freeze bypass invoked. Data-integrity and freeze-violation hard halts did not trigger. No row uses a calibration or learned strategy.

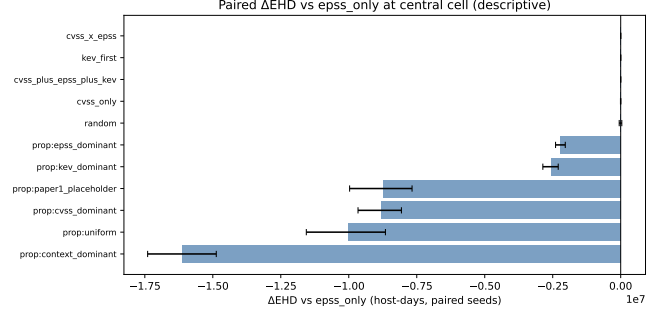


Figure 1. Paired-seed ΔEHD relative to `epss_only` per strategy (mean $\pm$ BCa 95% CI, 30 seeds). All baseline strategies (`cvss_only`, `kev_first`, `cvss_x_epss`, `cvss_plus_epss_plus_kev`) fall within the noise floor. Fixed-prior context-aware priors show larger descriptive differences but remain in the descriptive-only zone per the post-run variance criterion.

10.3. Operational EHD outcomes

The baseline `epss_only` central-cell mean EHD is $\approx 3.32 \times 10^9$ host-days (30 seeds). Paired-seed ΔEHD relative to `epss_only`, reported as descriptive observations:

- `cvss_only`: mean ≈ -163 hd, BCa CI $[-1,918, +1,329]$ — within noise floor; descriptive null.
- `kev_first`: mean $\approx +140$ hd, BCa CI $[-356, +839]$ — descriptive null.
- `cvss_x_epss`: mean $\approx +374$ hd, BCa CI $[-384, +1,145]$ — descriptive null.
- `cvss_plus_epss_plus_kev`: mean $\approx +103$ hd, BCa CI $[-224, +431]$ — descriptive null.
- `w_epss_dominant`: mean $\approx -2.21 \times 10^6$ hd, BCa CI does not overlap zero.
- `w_cvss_dominant`: mean $\approx -8.82 \times 10^6$ hd, BCa CI does not overlap zero.
- `w_kev_dominant`: similar order to `w_cvss_dominant`.
- `w_uniform`, `w_paper1_placeholder`: similar order to `w_epss_dominant`.
- `w_context_dominant`: mean $\approx -1.61 \times 10^7$ hd, BCa CI does not overlap zero.

These are descriptive observations relative to EPSS. Although several ΔEHD magnitudes exceed the 5,000-hd meaningful-effect threshold, the descriptive-language guardrail applies throughout: magnitudes are not interpreted as quality claims. No claim is made that any context-aware prior is meaningfully better than EPSS at this label sparsity (see post-run variance analysis below). Paired BCa CIs are the primary inferential quantity.

10.4. Weight-family sensitivity

Fifteen pairwise comparisons across the six fixed-prior vectors are recorded. Differences in mean EHD across vectors are observed but are not interpreted as

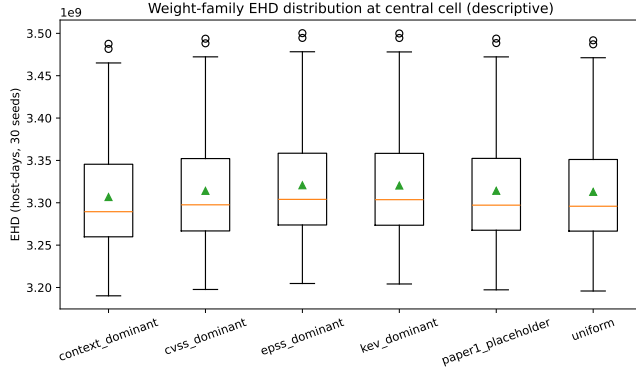


Figure 2. Weight-family sensitivity: Δ EHD across six fixed design-prior vectors relative to epss_only (mean $\pm$ BCa 95% CI, 30 seeds). Vector differences are descriptive observations; the post-run variance criterion demotes all axes to descriptive-only.

a quality ordering; the vectors are design priors, not evidence-based estimates.

10.5. Capacity sensitivity

Δ EHD across consecutive capacity steps $\{0.005 \rightarrow 0.01, 0.01 \rightarrow 0.02, 0.02 \rightarrow 0.05\}$ is reported for five strategy/vector arms. Records are populated with paired BCa CIs under the descriptive guardrail.

10.6. Blackout sensitivity

Blackout transitions $\{\text{none} \rightarrow \text{light} \rightarrow \text{primary} \rightarrow \text{strict}\}$ are evaluated for three strategies. No cell with zero scheduler feasibility occurred; the pre-registered floor of 0.95 was not approached.

10.7. Feature-ablation sensitivity

Twelve paired ablation-vs-full comparisons (two fixed-prior vectors $\times$ six feature ablations) are reported. Where the ablation-effect sign flips between vectors, both signs are recorded; no single vector is preferred.

10.8. Scheduler feasibility and audit integrity

`scheduler_feasibility_rate` = 1.0 on every primary cell. All four batches passed freeze-integrity verification; the Paper-1 freeze manifest SHA-256 was byte-equal before and after every batch.

10.9. Post-run variance analysis

Three post-run checks were recorded.

- Sensitivity measurability. Across all four axes (capacity, weight family, feature ablation, blackout), the absolute median paired Δ EHD is well above

the 1,000-hd null-axis threshold. Sensitivity is measurable at the cell-mean level.

- Catalog perturbation. One-product catalog perturbations were not executed at scale; this check is recorded as skipped and listed as a limitation.
- Within-cell variance criterion. The criterion fires on every primary robustness axis. The per-cell coefficient of variation across seeds ($CV_{\text{within}} \approx 0.024$) is approximately an order of magnitude larger than the across-cells coefficient (CV_{across}). Each axis is therefore demoted to descriptive-only: although cell-mean differences are large, per-cell seed noise is comparable to the between-cells signal at the 30-seed budget. No sensitivity claim survives this criterion; all results above are descriptive.

11. Discussion

Sparse public-feed labels are a methodological constraint, not a flaw in any particular model. The gate methodology is intended to surface that constraint before per-feature calibration is attempted, rather than discover it after weights have been fitted and presented. Treating the host-CVE record total as if it were the calibration unit is a misleading habit: the calibration unit is the CVE, and a few thousand records that share a handful of CVE-level labels do not provide more calibration information than the CVE count alone.

The sensitivity sweeps add reproducibility evidence and a set of axis-level descriptive observations: at the cell-mean level the priors do separate from each other, but at the seed level the per-cell variance is the same order of magnitude as the between-cells variance — the within-cell variance criterion fires on every axis. This is a useful signal in itself: it cautions against treating seed-aggregated mean EHD differences as decision evidence at this seed budget. A higher seed budget (deferred $n = 50$ and $n = 100$ runs) would allow this criterion to be re-evaluated.

The work says nothing about EPSS in general; the released v3-era EPSS is used as one feature among several, without retraining or critique. The work says nothing about whether KEV is an adequate ground-truth label in general; it is a sparse signal on the catalog $\times$ era examined. The work says nothing about CVSS quality. The work does not claim that context-aware priors are meaningfully better than EPSS for prioritization; descriptive observations that some priors produce lower EHD at the central cell are reported, and the sensitivity sweeps fall in the descriptive-only zone under the variance criterion.

We do not claim production-readiness. We do not claim deployment evidence. We do not claim government use. We do not claim compliance achievement.

The negative feasibility finding is useful precisely because it is honest and small. It is a data point about a specific catalog $\times$ era $\times$ label choice. It does not

generalise without further measurement, and we do not attempt that generalisation here.

12. Limitations

- Synthetic fleet. The fleet uses the Paper-1 synthetic generator; real-host generalisation is not claimed.
- 31-product catalog. Catalog choice is exogenous. A catalog-expansion study is required before the negative finding can be attributed to something other than catalog narrowness.
- CPE matching. The matching policy is `cpe_exact` at one configuration; fuzzy and manual matching were deferred.
- KEV-only labels. PoC and ExploitDB labels are license-gated and were not used. The negative finding reflects KEV-only label density at the specific catalog and era; richer label families may behave differently.
- Catalog perturbation. Catalog stability under one-product perturbations was not measured at scale; this remains an open question.
- EPSS v3 era only. The study is scoped to the EPSS v3 era, bounded by the model-version transition at 2025-03-17.
- Fixed priors are design priors. No vector is a published numeric estimate; the citation chains motivate the direction of each prior, not a specific magnitude.
- No calibration comparator. The calibration-infeasibility threshold prevents learned-weight cells; no fitted-weight comparator is included.
- Ranking-quality and KEV-deadline metrics are diagnostic-only due to label sparsity. They appear in appendix tables and do not enter the headline finding.
- Within-cell variance dominance. The post-run variance criterion fires on every sensitivity axis, demoting each to descriptive-only at the 30-seed budget; the seed budget is itself a limitation.
- No production-readiness. No production validation is claimed; the pipeline is not a production system.
- No deployment or compliance claims. No government-deployment or compliance claim is made.

13. Future Work

- Catalog expansion. Expand the product catalog and re-run the full gate, so the negative finding can be tested against catalog width.
- Per-t0 EPSS. The current study used a single cached EPSS snapshot at every t0 horizon. A future run with per-t0 EPSS snapshots would allow the EPSS feature to vary by horizon.

- Catalog perturbation. A one-product perturbation drift evaluation would quantify catalog stability under small changes.
- Larger seed budget. Sweeps at $n = 50$ and $n = 100$ seeds would allow the within-cell variance criterion to be re-evaluated and potentially yield inferential rather than descriptive sensitivity findings.
- Richer exploit labels. A licensed PoC / ExploitDB label path, under an explicit license-gate environment with no redistribution, would test whether label-family choice drives the sparsity finding.
- LEV integration. An evaluation incorporating NIST Likely Exploited Vulnerabilities scores [11] should be conducted once LEV scores stabilise.
- EPSS v4 era. A second gate measurement on the EPSS v4 era, treating v3 and v4 as separate slices, would test whether the negative finding persists across the model-version boundary.

14. Conclusion

This paper is a negative-result methodology paper. A pre-registered feasibility gate asked whether public-feed labels are dense enough for per-feature weight calibration on a public-sector-typical 31-product catalog over the EPSS v3 era. The gate measured only 7 unique positive distinct CVEs across 18 monthly horizons and 2,688 catalog-matched CVEs — well below the pre-registered threshold of 50. Calibration was not attempted. A pre-registered sensitivity sweep with six fixed design-prior vectors across capacity, blackout, and feature-ablation axes was executed; post-run variance analysis confirmed that within-cell seed variance dominates between-cell signal on every axis, so all sensitivity results are reported descriptively. The Paper-1 frozen artefact was preserved before and after every batch; the audit trail is content-addressed end-to-end. No claim is made that any prior is meaningfully better than EPSS, and no production-readiness is claimed. The reusable contribution is the failure-aware gate methodology plus the reproducible execution trail.

15. Reproducibility and Artifact Notes

All execution artifacts are content-addressed and available in the accompanying repository. The pre-registration document, stop-rule registry, and cell enumeration were locked before any data collection began.

Study design artifacts: pre-registration document, nine pre-registered fixes, minimal factorial cell enumeration (48 cells), batch plan, and stop-rule registry.

Gate measurement: multi-t0 feasibility probe outputs confirming 7 unique positive CVEs across 18 monthly horizons and 2,688 catalog-matched CVEs.

Pilot artifacts: pilot batch results, per-batch audit logs, and the pilot-gate decision record (PROCEED_TO_PRIMARY_NO_FALLBACK).

Primary artifacts: primary batch results across four batches (1,440 seed-runs, 48 cells, 8,640 metric rows), per-batch audit logs, and the primary-complete status record.

Aggregation outputs: summary tables for primary results, sensitivity sweeps (weight family, capacity, black-out, feature ablation), scheduler feasibility, and all five pre-registered inference families; figures for each axis.

The Paper-1 frozen artifact was not modified at any point during data collection or analysis; freeze integrity verification returns OK before and after every batch.

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HygieneBench: A Reproducible Synthetic Benchmark for Cyber-Hygiene Anomaly Detection Across Identity, Endpoint, and Patch Telemetry

Anonymous Author(s)

ABSTRACT

Cyber-hygiene anomaly detection—identifying stale privileged accounts, dormant account reactivations, endpoint coverage gaps, patch noncompliance clusters, and telemetry missingness—is operationally important but poorly served by existing public benchmarks, which focus predominantly on network intrusion or attack-emulation telemetry. We introduce HYGIENEBENCH, an open, reproducible benchmark that jointly covers Active Directory identity state, endpoint patch posture, vulnerability exposure, and telemetry freshness across seven evaluation tasks (T1–T7) and twelve anomaly classes. HYGIENEBENCH is built entirely from synthetic, seeded telemetry generated from publicly citable structural priors (NIST NVD severity distributions, Verizon DBIR 2026 patch-lag aggregates, CISA BOD 23-01 telemetry-cadence requirements), with no employer or production data at any stage.

We evaluate eight methods—a task-specific rule baseline (M1), a hybrid risk scorer (M2), Isolation Forest (M3), Local Outlier Factor (M4), One-Class SVM (M5), a linear autoencoder (M6), population-level temporal z-score (M7), and a graph-augmented Isolation Forest (M8)—across five telemetry conditions (baseline, fresh, stale, missing, unsupervised). Applying a pre-registered failure protocol ($\Delta < 0.05$ AP and $\Delta < 0.05$ P@k in $\geq 2/3$ seeds), we find that **86.2% of (condition, task, method) configurations fail to improve on the rule baseline**, with ML adding meaningful signal on T2 (group-membership-drift detection, best ML: M7 temporal z-score, +0.185 AP over rule) and T5 (patch-vulnerability hygiene, best ML: M5 OCSVM, +0.210 AP over rule). Telemetry staleness (C-STALE) consistently degrades detection performance relative to the baseline condition. We release the generator, datasets, and evaluation harness to enable reproducible comparison of future methods.

CCS CONCEPTS

• **Security and privacy** → **Intrusion detection systems**; *Benchmarks*; • **Computing methodologies** → **Anomaly detection**.

KEYWORDS

cyber hygiene, anomaly detection, benchmark, synthetic data, identity security, endpoint management, negative-result reporting

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1 INTRODUCTION

Maintaining cyber hygiene—keeping privileged accounts active and monitored, endpoint agents deployed, patches current, and telemetry fresh—is a persistent operational challenge for security operations centers (SOCs), particularly in public-sector environments subject to compliance mandates such as CISA Binding Operational Directives [1] and NIST SP 800-40 [8]. Failures in hygiene posture create the enabling conditions for identity-based attacks: dormant accounts reactivated by adversaries, endpoints missing agent coverage, and stale privileged credentials that may be invisible to behavioral analytics requiring recent baselines.

Despite the operational importance of hygiene-state monitoring, the anomaly-detection research community lacks a dedicated public benchmark for this problem. Existing benchmarks—the LANL Unified Host and Network Dataset [6], CERT Insider Threat datasets [5], CICIDS [11], and attack-emulation repositories such as Mordor/Security Datasets [9]—provide rich telemetry for network intrusion or insider-threat detection but do not model identity hygiene state, patch posture, vulnerability exposure, or telemetry freshness as first-class evaluation axes. Commercial platforms (Microsoft Defender for Identity, Splunk UBA, Exabeam) address related detection problems but are closed and do not release benchmark datasets.

This gap matters for two reasons. First, researchers who develop anomaly-detection methods for security operations have no way to compare their methods on hygiene-specific telemetry under controlled conditions. Second, there is no principled accounting of when unsupervised ML methods add value over simpler rule-based approaches for hygiene anomalies—a question with real operational implications, since rule maintenance is expensive and ML deployment requires justification.

This paper makes the following contributions:

- **C1: Synthetic generator.** A fully open, seeded synthetic telemetry generator covering eleven entity/event tables (Active Directory users, groups, computers, assets; login events; group membership events; endpoint patch state; vulnerability records; remediation events; account lifecycle events; telemetry freshness log) with documented structural priors.
- **C2: Anomaly taxonomy.** Twelve cyber-hygiene anomaly classes (AH-01 through AH-12) with ATT&CK enabling-condition mappings (not technique detection claims), injection protocols, and class imbalance specifications.

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- **C3: Benchmark task suite.** Seven evaluation tasks (T1–T7) under five telemetry conditions, with pre-registered k values, feature sets, and method applicability flags.
- **C4: Comparative evaluation, failure-aware.** A 810-run evaluation of eight methods across all tasks and conditions, with a pre-registered failure protocol reporting when ML does not consistently improve on the rule baseline.
- **C5: Negative-result reporting.** Explicit reporting and analysis of the 86.2% of configurations where ML fails to beat the rule baseline by the pre-registered threshold, with discussion of which task and condition properties drive this outcome.

The remainder of the paper is organized as follows. Section 2 reviews related benchmarks. Section 3 describes the HYGIENE BENCH design. Section 4 presents experimental setup. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

2 RELATED WORK

Network intrusion and attack emulation benchmarks. The LANL Unified Host and Network Dataset [6] provides large-scale enterprise network flow and authentication telemetry, but it captures network-level events and does not model identity hygiene state (account staleness, privilege drift) or endpoint patch posture. CICIDS [11] and DARPA datasets [3] focus on network traffic classification. Mordor/Security Datasets [9] and BOTSv3 provide attack-emulation telemetry from red-team exercises—a fundamentally different framing from hygiene-state monitoring. None of these benchmarks includes telemetry freshness or missingness as a controlled evaluation variable.

Insider threat datasets. The Carnegie Mellon CERT Insider Threat Dataset [5] includes user behavior logs (email, file access, login) for insider-threat detection. While it covers some identity-adjacent behaviors, it does not model endpoint patch state, vulnerability records, or the multi-table hygiene-state structure of enterprise AD environments.

Graph anomaly detection. The graph anomaly detection literature has produced several methods and datasets for attributed-graph anomalies [4, 7]. These approaches typically operate on single-hop graph structure and do not model the temporal, multi-table telemetry characteristic of enterprise security operations.

Commercial and proprietary platforms. Microsoft Defender for Identity, Entra ID Protection, Splunk UBA, Exabeam, and Microsoft Sentinel address overlapping detection problems. They are commercial products, not open benchmarks, and do not release datasets or evaluation methodologies for external comparison. We make no comparison claims against these platforms.

Synthetic security dataset generation. Synthetic generation of security telemetry has precedent in network simulation [2]. HYGIENE BENCH extends this tradition to the hygiene-state domain with documented structural priors and a reproducible seeded generator.

Failure-aware and negative-result reporting. The practice of explicitly reporting when ML fails to outperform simpler baselines is well-established in machine learning [10] but underrepresented

Table 1: HYGIENE BENCH v0.1 schema (medium scale, $n=1,000$ users).

Table	Rows	Description
users	1,000	AD user accounts with identity state
groups	65	AD security/distribution groups
computers	830	Managed endpoints
assets	830	Asset inventory entries
login_events	~249k	Auth. events (30-day window)
group_membership_events	~51	Group add/remove events
account_lifecycle_events	varies	Account create/disable/enable
endpoint_patch_state	830	Per-endpoint patch compliance
vulnerability_records	~3.5k	Per-endpoint CVE exposure
remediation_events	varies	Patch and remediation actions
telemetry_freshness_log	varies	Per-entity data-source freshness
anomaly_labels	110	Ground-truth labels (task+split)

in security benchmark papers. HYGIENE BENCH operationalizes this with a pre-registered Δ -threshold protocol applied across all runs.

3 HYGIENE BENCH DESIGN

3.1 Schema and Synthetic Generator

HYGIENE BENCH v0.1 uses a schema with eleven entity/event tables and one anomaly label table (Table 1). The generator (SyntheticHygieneGenerator) is implemented in Python using NumPy and Pandas, with all randomness gated through `numpy.random.default_rng(seed)`. Entity identifiers are deterministic UUIDs derived from `hashlib.md5(seed_tag + index)`, ensuring reproducibility across platforms.

Structural priors. Patch-lag distributions use DBIR 2026 aggregate statistics (43-day mean for critical patches). CVE severity distributions use NVD base score statistics. Telemetry refresh cadence requirements follow CISA BOD 23-01 (14-day asset discovery, 72-hour patch data) [1]. AD group-size distributions lack a peer-reviewed published source and are modeled via disclosed assumptions (power-law approximation) with sensitivity sweeps. All structural priors are documented in the dataset card.

Conditions. Five telemetry conditions parameterize freshness and missingness:

- **C-BASE:** No artificial staleness; all sources current.
- **C-FRESH:** Elevated freshness flagging; sources verified current.
- **C-STALE:** 20% of entities have heavy-stale telemetry (>14-day gap).
- **C-MISS:** One data source absent for 15% of entities.
- **C-UNSUP:** C-BASE with anomaly labels withheld from the training pipeline.

Table 2: HygieneBench anomaly classes. ATT&CK mappings are enabling-condition correlations; they do not imply detection of any specific technique.

Class	Code	ATT&CK condition	enabling	Tasks
Stale privileged account	AH-01	Valid Accounts: Do-main (T1078.002)		T1
Privilege escalation drift	AH-02	Valid Accounts; Do-main Policy Mod.		T7
Group membership drift	AH-03	Domain Policy Modification (T1484)		T2
Dormant account reactivation	AH-04	Valid Accounts: Do-main (T1078.002)		T6
Impossible/unusual login	AH-05	Valid Accounts; Lat-eral Movement		T1,T3
Endpoint-identity risk corr.	AH-06	Exploit Public-Facing App		T3
Patch noncompliance cluster	AH-07	Exploit Public-Facing App (T1190)		T5
KEV exposure aging	AH-08	Exploit Public-Facing App (T1190)		T5
Asset inventory mismatch	AH-09	Defense Evasion en-abling		T4
Missing endpoint agent	AH-10	Defense Evasion en-abling		T4
Telemetry missingness cluster	AH-11	Defense Evasion en-abling		T4
Abnormal remediation delay	AH-12	Exploit Public-Facing App		T5

Each condition is generated for three seeds (42, 137, 2024), producing 15 dataset instances (5 conditions $\times$ 3 seeds) with $n=1,000$ users per instance.

3.2 Anomaly Taxonomy

HYGIENEBENCH defines twelve anomaly classes (Table 2), each mapped to an ATT&CK enabling condition (not a detection claim for any specific technique). Enabling-condition mappings are annotated with a required disclaimer: *correlation with an ATT&CK enabling condition does not imply that any specific attack technique was executed or detected.*

3.3 Benchmark Tasks

Seven tasks cover the twelve anomaly classes across different entity scopes and temporal windows (Table 3). Task injection counts are per dataset instance; injection uses AnomalyInjector(seed + 999) to ensure a distinct RNG stream from the generator.

Dataset splits. Labels are stratified 60/20/20 by anomaly class. Anomalous entities are assigned to the test split, with normal entities split randomly across train/val/test. The split is deterministic given the seed.

4 EXPERIMENTAL SETUP

4.1 Methods

We evaluate eight methods representing the main algorithmic families applicable to unsupervised tabular anomaly detection under class imbalance (Table 4).

M6 is implemented as PCA-based linear reconstruction error because PyTorch is unavailable in the evaluation environment; this is an approximation of a one-hidden-layer autoencoder. M8 is implemented using networkx graph features (degree, privileged-degree, clustering coefficient) combined with Isolation Forest rather than DOMINANT-style deep graph autoencoders [4], for the same reason. Both approximations are documented. M8 is excluded from T4 and T5 by design (no meaningful graph signal for asset/patch anomalies; reported as N/A).

4.2 Feature Engineering

For each task, TaskFeatureExtractor.extract(task_id, seed) computes a feature matrix from raw CSVs. Features include task-relevant identity state attributes, temporal aggregates (e.g., 30-day login frequency, off-hours rate, cross-segment rate), patch compliance scores, vulnerability counts, KEV exposure counts, and telemetry freshness scores. Missing values are imputed with training-set column means before fitting.

4.3 Metrics and Negative-Result Protocol

Primary metrics. Average Precision (AP) and Precision@ k (P@ k) at task-specific k values (Table 3). AP is the standard interpolated area under the precision-recall curve. P@ k is the fraction of true anomalies in the top- k ranked entities—the operationally relevant metric for SOC review budgets.

Secondary metrics. Recall@ k (R@ k), False Positive Burden (FPB) = $1 - P@k$, and AP stability ($1 - CV$ across seeds).

Pre-registered failure protocol. A (condition, task, method) configuration is flagged as a *negative result* ($\blacktriangleright$) if:

$$AP(\text{method}) - AP(M1) < 0.05 \text{ and } P@k(\text{method}) - P@k(M1) < 0.05 \quad (1)$$

in at least $\lceil 2/3 \times n_{\text{seeds}} \rceil = 2$ of 3 seeds. Flagged configurations are reported in full and are not filtered. This protocol was pre-registered prior to examining any results.

5 RESULTS

5.1 Overall Performance

Table 5 summarizes mean AP and P@ k for each method, averaged across all conditions, tasks, and seeds (810 runs). The rule baseline (M1) achieves the highest mean AP (0.916). M5 (OCSVM) is the closest ML competitor (AP = 0.913, P@ k = 0.447 vs. M1 P@ k = 0.444). M2 (hybrid scorer) performs worst among all methods (AP = 0.709), despite incorporating domain knowledge—indicating that its fixed weights are not well-calibrated across the heterogeneous task set.

Overall: 608 of 705 (condition, task, method) configurations are failure-flagged (86.2% negative-result rate). For synthetic cyber-hygiene telemetry with structured, rule-exploitable anomaly injection, simple threshold rules perform at least as well as unsupervised ML methods on the majority of evaluation

Table 3: Benchmark tasks. k values are pre-registered per TASK_SPECS.md and reflect operational SOC review budgets.

Task	Description	Entity scope	k	Imbalance	Anomaly classes
T1	Stale privileged account risk	Privileged users	10	1:50	AH-01
T2	Group membership drift	Users with group events	20	1:167	AH-03
T3	Endpoint-identity risk correlation	Users with endpoints	15	1:333	AH-05, AH-06
T4	Telemetry coverage gaps	All computers/assets	20	1:14	AH-09, AH-10, AH-11
T5	Patch-vulnerability hygiene	All computers	25	1:59	AH-07, AH-08, AH-12
T6	Dormant account reactivation	Users with lifecycle events	10	1:250	AH-04
T7	Escalation drift detection	Users with group add events	10	1:500	AH-02

Table 4: Evaluation methods. M8 is excluded from T4 and T5 (no graph signal; reported as N/A).

ID	Name	Type	Notes
M1	Rule baseline	Task-specific rules	No training; per-task thresholds
M2	Hybrid scorer	Weighted combination	Identity + endpoint + vuln + freshness
M3	Isolation Forest	Ensemble	200 trees, contam.=0.02
M4	LOF	Density-based	$k=20$ neighbors, novelty mode
M5	OCSVM	Kernel-based	$\nu=0.05$, RBF kernel
M6	Linear-AE	Reconstruction error	PCA (latent dim=8); approx. neural AE
M7	Temporal z-score	Statistical	Population z-score, temporal features
M8	Graph-IF	Graph + ensemble	Bipartite features + IF; approx. DOMINANT

Table 5: Mean AP and P@ k by method across all 810 runs. ► = failure-flagged (ML does not improve on rule by $\Delta=0.05$ in $\geq 2/3$ seeds). †M8 covers T1–T3, T6–T7 only ($n=75$ configs).

Method		Mean AP	Mean P@ k	Failure rate
M1	Rule baseline	0.916	0.444	—
M5	OCSVM	0.913	0.447	69.5% ►
M8	Graph-IF†	0.801	0.348	100% ►
M4	LOF	0.800	0.445	77.1% ►
M6	Linear-AE	0.798	0.407	88.6% ►
M3	Isolation Forest	0.781	0.428	88.6% ►
M7	Temporal z-score	0.774	0.392	87.6% ►
M2	Hybrid scorer	0.709	0.417	96.2% ►
Overall failure rate: 608/705 = 86.2%				

configurations. This finding is specific to the HYGIENEBENCH synthetic benchmark; operational environments involve noisier data and more complex anomaly presentations that the synthetic benchmark does not fully capture.

5.2 Where ML Adds Value

Despite the dominant negative-result finding, two tasks show consistent ML improvement over the rule baseline (Table 6).

T2—Group Membership Drift. The rule baseline (M1) achieves mean AP = 0.766 on C-BASE and C-STALE. The best ML method

Table 6: Tasks where ML consistently adds value ($\Delta AP > 0.05$ in $\geq 2/3$ seeds).

Task	Best ML	Rule AP	ML AP	ΔAP	Condition
T2	M7 (Z-score)	0.766	0.951	+0.185	C-BASE/STALE
T2	M4 (LOF)	0.855	0.975	+0.120	C-MISS
T5	M5 (OCSVM)	0.458	0.668	+0.210	C-BASE/STALE
T5	M5 (OCSVM)	0.726	0.837	+0.111	C-MISS

is M7 (temporal z-score), achieving AP = 0.951 (+0.185 AP). M5 (OCSVM) is second at AP = 0.913 (+0.147) and M4 (LOF) third at AP = 0.890 (+0.124). Three ML methods consistently improve on the rule baseline. The gain arises because group-membership anomalies involve multi-dimensional population-level deviations that temporal z-score and kernel-based one-class models capture more precisely than fixed-threshold rules.

T5—Patch-Vulnerability Hygiene. This is the hardest task overall (mean AP = 0.617 across all methods). The rule baseline achieves AP = 0.458 on C-BASE/C-STALE, while the best ML method achieves AP = 0.668 (+0.210 AP) via M5 (OCSVM). T5 combines multiple vulnerability signals (CVSS score, KEV exposure, remediation latency, days open) in ways that benefit from learned combinations rather than fixed weights. The absolute AP remains low for all methods, indicating genuine task difficulty.

5.3 Where Rules Dominate

Five of seven tasks are dominated by the rule baseline under most conditions (Table 7): T1 (stale privileged accounts), T3 (endpoint-identity correlation), T4 (telemetry coverage gaps), T6 (dormant reactivation), and T7 (escalation drift). For these tasks under C-BASE, both M1 and the best ML method achieve AP $\approx$ 1.000.

T3 is a partial exception. Under C-BASE its AP is 0.782, but under C-STALE it degrades sharply to 0.613 ($\Delta = -0.168$)—the largest condition sensitivity of any task. T3 combines identity-side and endpoint-side features, making it susceptible to staleness in either domain.

T7 exhibits a ceiling effect: with only 2 anomalous entities per dataset instance (imbalance $\approx$ 1:500), all methods achieve AP = 1.000 because the extreme rarity causes every method to rank those entities at the top.

Table 7: Rule-dominated tasks (rule AP $\geq$ best-ML AP in majority of configs).

Task	Mean AP	Notes
T1 Stale priv. accounts	0.937	AP $\approx$ 1.0 under all conditions
T3 Endpoint-identity	0.736	C-STALE sensitivity: -0.168
T4 Telemetry gaps	0.890	High AP; 1:14 imbalance
T6 Dormant reactivation	0.832	AP $\approx$ 1.0; M8 exception (0.490)
T7 Escalation drift	0.822	Extreme imbalance; ceiling effect

Table 8: Mean AP by telemetry condition (all methods and tasks).

Condition	Mean AP	Δ vs BASE	Primary driver
C-FRESH	0.856	+0.084	T5 +0.238; T1 +0.101
C-MISS	0.845	+0.073	T5, T1 dominate
C-UNSUP	0.844	+0.072	Labels withheld
C-BASE	0.772	—	Reference
C-STALE	0.741	-0.031	T3 -0.168 ; staleness

5.4 Effect of Telemetry Conditions

Table 8 shows mean AP by condition. C-STALE produces the lowest mean AP (0.741 vs. C-BASE 0.772), confirming that staleness degrades detection.

The C-FRESH, C-MISS, and C-UNSUP conditions show higher mean AP than C-BASE (0.856, 0.845, 0.844). The largest contributor is T5 (+0.238 AP under C-FRESH). Fresh telemetry improves patch compliance scores, KEV exposure counts, and remediation latency reliability, making the underlying feature signals more discriminating. This is a data-quality effect, not a freshness-flag artifact.

5.5 Failure-Flag Summary

Figure 1 shows the failure-flag heatmap. Key patterns:

- **M8 (graph) is failure-flagged on 100% of configs (75/75).** M8 produces non-trivial anomaly scores (e.g., T1 AP = 0.919, T3 AP = 1.000) but fails to improve on the rule baseline on any configuration. The largest deficit is T6 (dormant reactivation): M8 AP = 0.490 vs. M1 AP = 1.000. Bipartite graph features capture structural position, not temporal account state.
- **M5 (OCSVM) has the lowest failure rate (69.5%)** and is not flagged on T2 and T5, the two tasks where ML adds value.
- **M2 (hybrid scorer) is failure-flagged on 96.2% of configs—**worst among non-graph methods. Its fixed-weight combination is dominated by the task-specific rule baseline.

6 DISCUSSION

6.1 Interpretation of the Negative-Result Finding

The 86.2% failure rate is not a finding against ML in security operations broadly. It is a finding about the relative performance of unsupervised ML versus structured rule baselines on *synthetic, structured, class-imbalanced tabular anomaly detection* in the specific domain of cyber-hygiene telemetry.

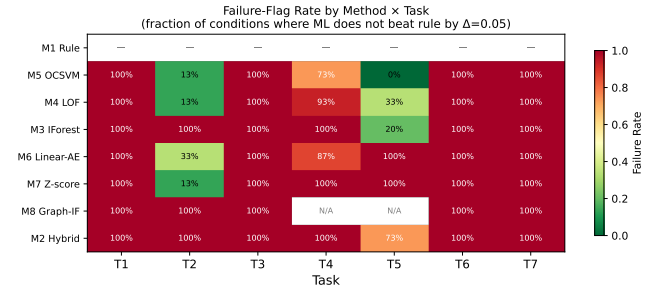


Figure 1: Failure-flag rate heatmap (method $\times$ task). Dark cells indicate configurations where ML fails to improve on the rule baseline by the pre-registered $\Delta=0.05$ threshold in $\geq 2/3$ seeds.

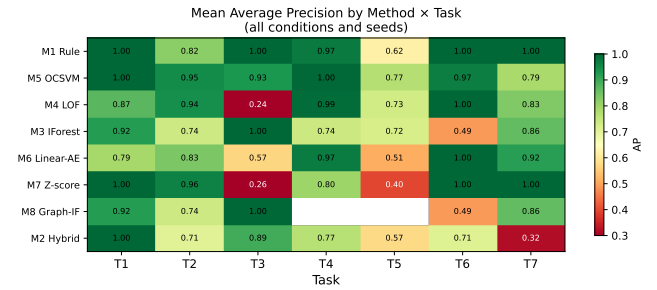


Figure 2: Condition-averaged AP heatmap (method $\times$ task). Higher values indicate better anomaly ranking. M1 (rule baseline) anchors the top row.

The synthetic benchmark is, by design, favorable to rule-based methods: anomaly injection creates features that are directly correlated with the anomaly class in ways that rule thresholds exploit cleanly. In operational environments, data is noisier, labels are unavailable, and the feature space is more complex. HYGIENEBENCH reports this limitation explicitly in the dataset card and generator documentation.

The appropriate takeaway is: for HYGIENEBENCH v0.1, ML adds consistent value over rule baselines specifically on T2 (group membership drift) and T5 (patch/vulnerability hygiene). These tasks involve multi-signal combinations that benefit from learned representations. Future work should test whether this advantage persists under increased population-level noise and label sparsity.

6.2 Limitations

Synthetic data. All telemetry is synthetic. Structural priors are approximate (DBIR 2026 for patch lag; NVD for CVE severity; CISA BOD 23-01 for freshness cadence). AD group-size distributions lack a peer-reviewed source and are modeled under disclosed assumptions. Detection performance on real operational data will differ from benchmark results.

Approximations in M6 and M8. M6 is implemented as PCA-based linear reconstruction error (not a neural autoencoder) and M8 as networkx graph features + Isolation Forest (not DOMINANT-style

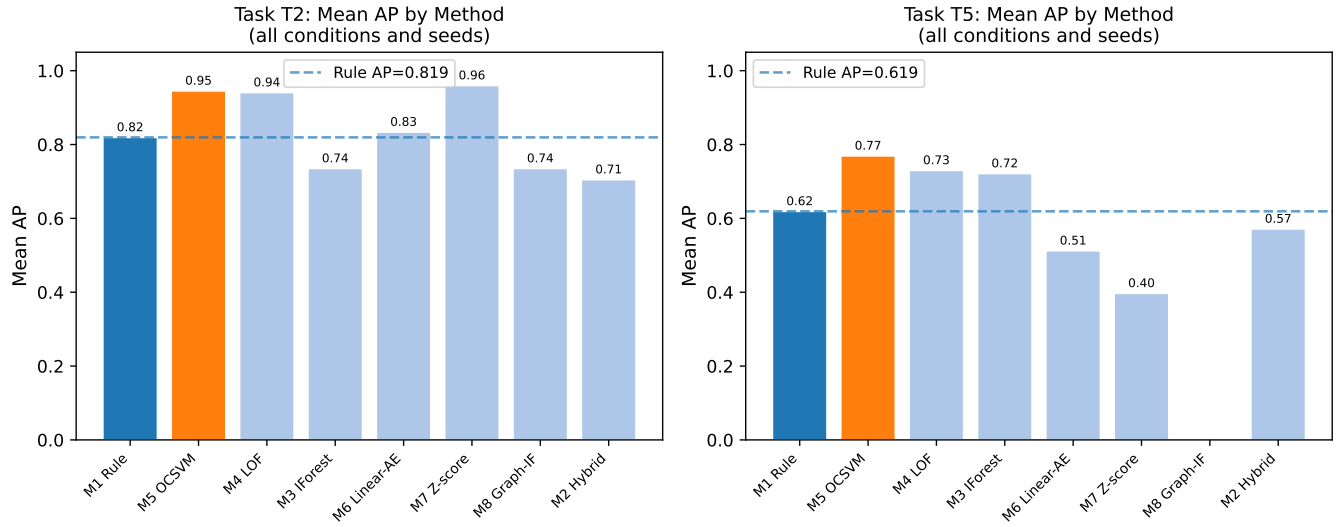


Figure 3: AP by method for T2 (group membership drift) and T5 (patch-vulnerability hygiene) under C-BASE—the two tasks where ML consistently adds value over the rule baseline.

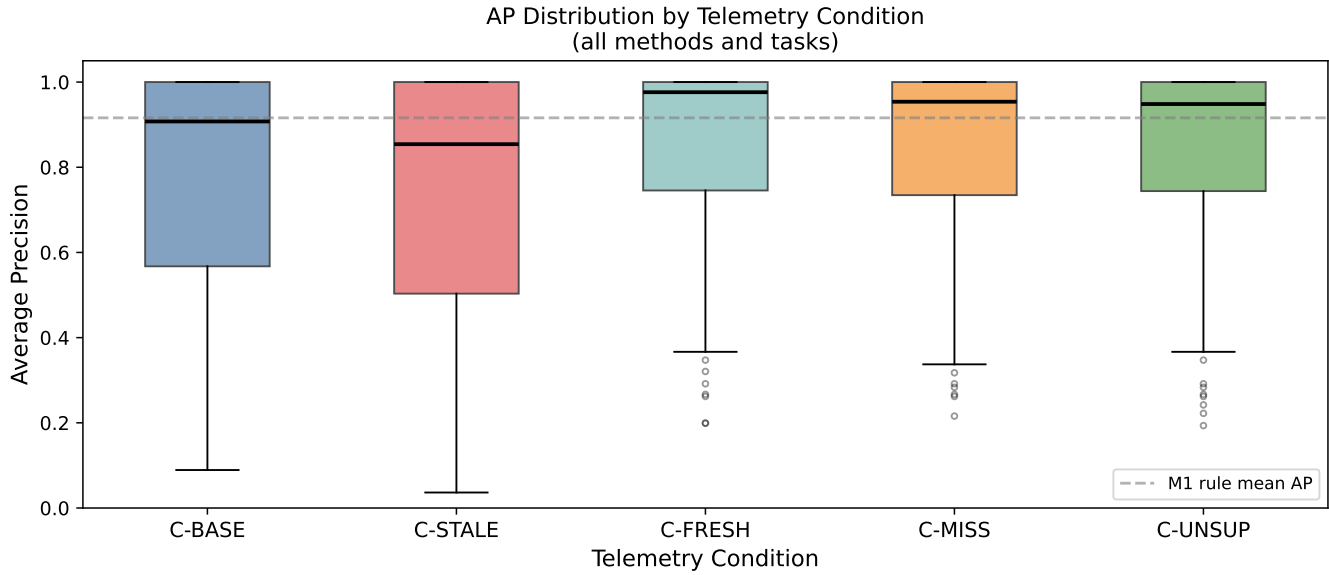


Figure 4: AP distribution by telemetry condition (all tasks and methods). C-STALE has the lowest mean AP; C-FRESH has the highest.

deep graph autoencoder [4]). These are disclosed approximations; results may change with full implementations.

Small population scale. Medium scale ($n=1,000$ users) limits the statistical power of the evaluation. The population is smaller than most public-sector environments, which may affect the generalizability of the class imbalance findings.

Feature engineering as a design choice. The task feature extractors (T1–T7) encode domain knowledge about which features matter. Methods that exploit these features more effectively will appear better. A fully unsupervised method that discovers relevant features from raw tables is not evaluated.

Fixed k values. $P@k$ is sensitive to the choice of k . HYGIEN-BENCH uses operationally motivated k values (daily/weekly SOC

review budgets) rather than tuning k to maximize method performance. This is a deliberate design choice but limits comparison to benchmarks with different k conventions.

No calibration. Anomaly scores are not calibrated to probabilities. Calibration is the primary research question of a separate line of work and is excluded from HYGIENE BENCH by design.

6.3 Relationship to Prior Work

HYGIENE BENCH is designed to complement, not replace, existing benchmarks. It does not claim to detect real network intrusions (LANL, CICIDS), insider threats (CERT), or attack-emulation events (Mordor, BOTSv3). It is the first benchmark, to our knowledge, that jointly covers Active Directory identity hygiene state, endpoint patch posture, vulnerability exposure, and telemetry freshness under controlled conditions with an openly released, reproducible synthetic generator.

7 CONCLUSION

We introduced HYGIENE BENCH, a reproducible synthetic benchmark for cyber-hygiene anomaly detection, covering seven tasks and twelve anomaly classes across eleven entity/event tables. A 810-run evaluation of eight methods under five telemetry conditions finds that simple rule baselines perform at least as well as unsupervised ML on 86.2% of (condition, task, method) configurations. ML adds meaningful value on two of seven tasks—group membership drift (T2, +0.185 AP) and patch/vulnerability hygiene (T5, +0.210 AP)—while rule baselines dominate on tasks with strong single-feature discriminators. Telemetry staleness (C-STALE) consistently degrades detection performance. These findings motivate honest, task-stratified evaluation of anomaly detection methods and explicit negative-result reporting in security benchmark papers.

HYGIENE BENCH v0.1, including the synthetic generator, frozen datasets, and evaluation harness, is openly available at: <https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper3> — Zenodo DOI to be finalized at camera-ready submission.

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HygienePrio: Cyber-Hygiene Signal Augmentation for EPSS-Weighted Vulnerability Prioritization

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Abstract—Capacity-constrained vulnerability remediation teams must decide which (host, CVE) pairs to patch within each maintenance window, yet the dominant public triage signal — the Exploit Prediction Scoring System (EPSS) — is asset-agnostic by design: it estimates per-CVE exploit likelihood without regard to whether the affected host is actively managed, has high patch debt, or is exposed through privileged Active Directory (AD) relationships. Prior work found that augmenting EPSS with CVE-level features did not improve prioritization on a synthetic government-shaped fleet, suggesting that the missing signal is dynamic host-level hygiene posture rather than additional CVE-level features.

We present HygienePrio, a scoring framework that integrates a three-dimensional Hygiene Risk Score (HRS) — capturing patch posture, AD exposure state, and telemetry freshness — into an EPSS-weighted (host, CVE) pair prioritization scorer with an explicit interaction term. Weights are calibrated by pre-registered grid search on held-out fleet seeds.

In a pre-registered synthetic evaluation across 25 fleet seeds, HygienePrio achieves a mean Precision@50 of 0.509 compared with EPSS-only’s 0.202 — approximately 31 percentage-point improvement under capacity-constrained schedules, with non-overlapping BCa 95% confidence intervals and improvement on all 25 of 25 evaluation seeds (synthetic evaluation only). Patch posture is the dominant hygiene dimension (≈ 20 pp ablation drop at P@50), substantially exceeding the contributions of AD exposure (≈ 9 pp) and telemetry freshness (≈ 2 pp, overlapping confidence intervals).

To our knowledge, HygienePrio is the first scorer to jointly operationalize per-CVE exploit likelihood and host-level hygiene posture for remediation ordering in a pre-registered, reproducible evaluation framework. All claims are bounded to the synthetic evaluation context; generalization to real enterprise fleets requires external validation on real exploitation outcome data not performed here.

Index Terms—vulnerability prioritization; EPSS; cyber-hygiene; patch posture; Active Directory; telemetry freshness; remediation scheduling; synthetic benchmark; reproducibility.

I. Introduction

Public-sector and enterprise endpoint fleets accumulate disclosed vulnerabilities faster than operations teams can remediate them within constrained maintenance windows. The practical triage question is not which vulnerabilities are inherently severe but which (host, CVE) pairs should receive the next unit of limited remediation capacity — a distinction that requires combining exploit-likelihood signals with knowledge of which hosts are most exposed and least well-maintained.

The Exploit Prediction Scoring System (EPSS) [1] has emerged as the leading public signal for per-CVE exploit likelihood, outperforming CVSS base scores as a predictor

of near-term exploitation in the wild [2]. EPSS is incorporated into risk-based vulnerability management (RBVM) tools, CISA vulnerability prioritization guidance [3], [4], and remediation policy frameworks. However, EPSS is asset-agnostic by design: it assigns a single probability score to a CVE regardless of whether the vulnerable host is a freshly-telemetered, heavily-patched workstation or an unmanaged endpoint with stale patch data and over-privileged AD group memberships.

Intuitively, a CVE with moderate EPSS on a host that has gone 21 days without telemetry, carries 40% unpatched CVE debt, and is accessible from domain-privileged accounts represents a materially different risk than the same CVE on a well-maintained host with fresh telemetry and minimal AD exposure. Yet EPSS rankings treat both identically. Host-level hygiene posture — the aggregate state of patch compliance, identity and access management hygiene, and telemetry coverage — is a natural complement to exploit-likelihood scoring.

Prior work directly relevant to this problem is VulnPrio [5] (referred to throughout as Paper 1), which built a reproducible benchmark for (host, CVE) pair prioritization on a synthetic government-shaped fleet (EEHDA) and evaluated 13 strategies combining EPSS, CVSS, KEV membership, asset criticality, and endpoint exposure signals. Paper 1’s central null finding was that the proposed multi-factor model was statistically indistinguishable from the EPSS-only baseline: inter-strategy differences fell within seed-to-seed variation, and no CVE-level augmentation strategy materially outperformed EPSS. Paper 1 concluded that this result exposed capacity-driven and base-rate trade-offs and provided a falsifiable benchmark on which calibrated studies could build — explicitly framing the result as a foundation for future work rather than an endpoint.

Separately, HygieneBench [6] (Paper 3) introduced a reproducible benchmark for cyber-hygiene anomaly detection on a synthetic fleet covering Active Directory identity state, endpoint patch posture, vulnerability exposure, and telemetry freshness. HygieneBench’s primary finding was that 86.2% of evaluated (condition, task, method) configurations fail to outperform a rule baseline, but that patch-vulnerability hygiene (Task T5, +0.210 average precision over rule for OCSVM) and group-membership drift (Task T2, +0.185 AP for temporal z-score) are exceptions where structured ML methods add meaningful signal.

The central hypothesis of this paper is that the sig-

nal missing from Paper 1’s multi-factor model was not additional CVE-level features but host-level hygiene posture: patch debt, AD privilege exposure, and telemetry freshness. HygieneBench demonstrated that these dimensions carry discriminative structure detectable from the synthetic fleet telemetry. HygienePrio operationalizes this hypothesis by constructing a Hygiene Risk Score (HRS) from HygieneBench-style host telemetry and integrating it into an EPSS-weighted (host, CVE) scorer via an additive combination with an explicit interaction term.

A. Contributions

This paper makes the following contributions:

- C1 — HygienePrio scorer: A four-component weighted scoring function $S(h, c) = \alpha \times \text{EPSS}(c) + \beta \times \text{HRS}(h) + \gamma \times \text{KEV_recency}(c) + \delta \times (\text{EPSS}(c) \times \text{HRS}(h))$ that integrates host-level hygiene posture with per-CVE exploit likelihood in a principled, interpretable form.
- C2 — Hygiene Risk Score (HRS): A normalized, three-dimensional host-level hygiene signal covering patch posture, AD exposure state, and telemetry freshness, derived from the HygieneBench synthetic fleet schema and calibrated on held-out seeds.
- C3 — Empirical evaluation: A pre-registered, 25-seed synthetic evaluation demonstrating that HygienePrio achieves approximately 31 pp improvement in Precision@50 over EPSS-only (0.509 vs. 0.202; non-overlapping BCa 95% CIs; synthetic evaluation), with patch posture as the dominant HRS dimension (~ 20 pp ablation drop) and AD exposure as a substantial secondary contributor (~ 9 pp).
- C4 — Failure mode analysis and operationalization boundary: An explicit identification of conditions under which HygienePrio’s advantage is smaller (interaction term equivocal; freshness dimension marginal) and the structural boundary conditions (ground truth circularity, synthetic data, 5-seed calibration) that bound the positive finding. This analysis prevents over-generalization and frames the result as a pre-clinical finding rather than a deployment recommendation.

B. Paper Organization

Section 2 reviews background and related work. Section 3 formalizes the problem. Section 4 describes the synthetic dataset and fleet. Section 5 presents the HygienePrio methodology. Section 6 describes the experimental setup. Section 7 reports results. Section 8 discusses findings, limitations, and connections to prior papers. Section 9 addresses threats to validity. Section 11 concludes.

II. Background and Related Work

A. EPSS and Exploit-Likelihood-Based Prioritization

The Exploit Prediction Scoring System (EPSS) [1] is a machine-learning model that estimates the probability

that a published CVE will be exploited in the wild within the next 30 days. EPSS v3 uses $\sim 1,500$ features drawn from CVE descriptors, NVD metadata, and threat intelligence feeds, and produces daily probability scores for all published CVEs. Empirical evaluations [2] have shown EPSS substantially outperforms CVSS base scores as a predictor of near-term exploitation: CVEs in the top decile of EPSS account for a disproportionate fraction of observed exploitation events. EPSS is now integrated into CISA’s vulnerability prioritization guidance and is used by numerous RBVM vendors.

CISA’s Known Exploited Vulnerabilities (KEV) catalog [7] is a complementary signal: it lists CVEs confirmed to have been exploited in the wild, with mandated remediation deadlines for federal agencies under BOD 22-01 [3]. KEV membership provides strong prioritization signal but covers only a small fraction of the CVE backlog ($\sim 15,000$ of the 200,000+ published CVEs as of 2026). Prior work has studied optimal combined use of EPSS and KEV [5], generally finding that KEV provides a high-precision but low-recall prioritization set, while EPSS provides broader coverage at reduced precision.

The asset-agnostic limitation. A structural limitation of both EPSS and KEV is that they are CVE-level signals: they score a vulnerability, not a (vulnerability, host) pair. For remediation scheduling, the unit of action is the pair: patching CVE-X on Host-A is a distinct action from patching CVE-X on Host-B, with different operational cost, risk reduction, and scheduling feasibility. EPSS provides no signal about which host carrying a given CVE should be remediated first.

B. CVE-Level Host-Context Augmentation

To our knowledge, no published academic work has systematically incorporated dynamic host-level hygiene posture (patch debt, identity exposure, telemetry freshness) into EPSS-based (host, CVE) pair prioritization. Asset criticality models weight CVE severity by static host-role classifications, producing host-adjusted scores. Exposure models incorporate network reachability, service exposure, and lateral movement risk. Commercial RBVM tools (Tenable Lumin, Qualys TruRisk, Rapid7 InsightVM) combine such signals proprietarily, but their scoring methods are not published and results are not independently reproducible.

Paper 1 (VulnPrio) [5] is the closest public predecessor to HygienePrio. It built an open, reproducible benchmark for (host, CVE) pair prioritization on a synthetic government fleet (EEHDA), evaluating 13 strategies combining EPSS, CVSS, KEV, asset criticality, endpoint exposure, and multi-factor composite scores across 30 seeds. The central finding was null: the proposed multi-factor model was statistically indistinguishable from EPSS-only on the primary metric (expected exploited host-days, EHD), with all inter-strategy differences falling within seed-to-seed variation. Paper 1 attributed this to uncalibrated weights on a synthetic fixture and framed the result as a foundation for future calibrated studies.

Differentiating HygienePrio from Paper 1. Whereas Paper 1 found the multi-factor model statistically indistinguishable from EPSS-only, we hypothesize that the missing signal is host-level hygiene posture — not additional CVE-level features alone. Paper 1’s augmentation features (asset criticality, endpoint exposure, KEV flags) are all CVE-level or static host-role features. HygienePrio adds dynamic, telemetry-derived hygiene signals: how much patch debt is currently outstanding on this specific host, how exposed this host is through AD group memberships and privileged accounts, and how recently this host has reported telemetry. These are fundamentally different in kind from asset criticality scores and network exposure flags.

C. Cyber-Hygiene Benchmarks and Anomaly Detection

HygieneBench (Paper 3) [6] introduced the first open benchmark for cyber-hygiene anomaly detection covering AD identity state, endpoint patch posture, vulnerability exposure, and telemetry freshness. Its evaluation of eight detection methods across 12 anomaly classes and seven tasks found that 86.2% of (condition, task, method) configurations fail to outperform a rule baseline, but identified task T5 (patch-vulnerability hygiene) and T2 (group-membership drift) as exception zones where structured methods add meaningful signal. HygienePrio draws on the HygieneBench schema as the source of HRS features and inherits its fleet generation logic.

Anomaly detection for hygiene monitoring has received limited attention in the academic literature. Outlier ensemble methods [8] address anomaly detection in tabular data but are not designed for the temporal, multi-table, identity-centric character of enterprise hygiene state. To our knowledge, no existing public benchmark simultaneously models AD identity state, endpoint patch posture, vulnerability exposure, and telemetry freshness as first-class evaluation axes; adjacent benchmarks are surveyed in [6].

D. Patch Management and Remediation Scheduling

NIST SP 800-40 Rev. 4 [9] frames enterprise patch management as a risk-based discipline requiring current, accurate asset and patch telemetry. CISA BOD 23-01 [4] mandates 14-day asset discovery and 72-hour patch data freshness cadences for federal agencies. These mandates implicitly encode hygiene requirements: fleets that fail them have stale telemetry, reducing the reliability of any signal-based prioritization. HygienePrio models this explicitly through the TelemetryFreshness dimension of HRS.

Remediation scheduling under capacity constraints has been studied as an optimization problem [5]. The key insight from this literature is that prioritization under bounded capacity is a truncation problem: rank quality at the top- K positions matters more than rank quality across the full list, because capacity constraints mean that pairs outside the top- K are never acted on regardless of

their rank. This motivates Precision@ K as the primary evaluation metric.

III. Problem Formulation

A. Setting

Let F denote a fleet of hosts $H = \{h_1, \dots, h_N\}$ and let $C = \{c_1, \dots, c_M\}$ denote a set of disclosed CVEs. Let $V \subseteq H \times C$ be the set of applicable (host, CVE) pairs: $(h, c) \in V$ if CVE c is applicable to host h (i.e., host h runs vulnerable software affected by c). The set V is derived from the patch state and vulnerability records tables.

Let K be the remediation capacity: the number of (host, CVE) pairs that a team can remediate within a single maintenance window. A prioritization method produces a ranked list R of $|V|$ pairs; the top K elements of R define the action set for the window.

B. Ground Truth

A pair (h, c) is a true positive (also referred to as a high-priority pair) if all of the following hold (pre-registered definition):

- 1) $\text{EPSS}(c) > 0.10$ — the CVE has non-trivial exploit likelihood. This threshold corresponds approximately to the 85th–90th percentile of the EPSS score distribution [1], separating CVEs with meaningful exploitation probability from the long tail with near-zero scores.
- 2) $\text{HRS}(h) > \text{fleet 75th percentile}$ — the host is in the bottom quartile of hygiene posture. The 75th percentile is chosen as a pre-registered conservative boundary that identifies genuine hygiene outliers while avoiding sensitivity to the exact threshold value for the large majority of pairs.
- 3) $(h, c) \in V$ — the CVE is applicable to the host.

Condition 1 ensures the CVE is at meaningful risk of exploitation. Condition 2 ensures the host is a hygiene outlier rather than a well-managed endpoint. The conjunction ensures HygienePrio’s design goal — identifying high-exploit-likelihood CVEs on hygiene-poor hosts — has an operational ground truth against which to evaluate.

Note on circularity: The use of HRS in both the scorer (via $\text{HRS}(h)$) and the ground truth (via the $\text{HRS} > 75\text{th percentile}$ condition) creates a structural advantage for HygienePrio over baselines that do not use HRS. This is acknowledged and discussed in §9 (Threats to Validity). The ground truth definition is pre-registered and fixed; it is not modified post-hoc to favor any method.

C. Metrics

Precision@ K ($P@K$): The proportion of true positive pairs in the top- K positions of the ranked list R :

$$P@K = \frac{|\{(h, c) \in \text{top}_K(R) : (h, c) \text{ is true positive}\}|}{K}$$

Evaluated at $K = 50, 100, 250$.

TABLE I

EEHDA table schema and role in HygienePrio (medium-scale fleet)

Table	Rows	Role in HygienePrio
users	1,000	AD user account attributes
computers/assets	830	Host inventory; maps hosts to users
endpoint_patch_state	830	Per-host patch compliance snapshot (PatchPosture source)
vulnerability_records	~3,500	Per-host CVE exposure (V, EPSS scores)
telemetry_freshness_log	varies	Per-host data freshness (TelemetryFreshness source)
groups/group_membership_events	65/~51	AD group membership (ADExposure source)
anomaly_labels	110	HygieneBench ground truth (informational only)

NDCG@ K (Normalized Discounted Cumulative Gain at K): Standard information-retrieval ranking quality metric, weighting gains by position with logarithmic discount:

$$\text{NDCG@}K = \frac{\text{DCG@}K}{\text{IDCG@}K}$$

where $\text{DCG@}K = \sum_{i=1}^K \text{rel}_i / \log_2(i+1)$, $\text{rel}_i \in \{0, 1\}$ is the relevance of the i -th ranked pair, and $\text{IDCG@}K$ is the DCG of the ideal ranking. Evaluated at $K = 50, 100, 250$.

Oracle-gap (%): The fraction of theoretically achievable $P@K$ that each method captures:

$$\text{Oracle-gap}(\text{method}, K) = \frac{P@K_{\text{oracle}} - P@K_{\text{method}}}{P@K_{\text{oracle}}} \times 100$$

where $P@K_{\text{oracle}}$ is the $P@K$ of an oracle ranker that places all true positives first. Oracle-gap provides a normalized view of how far each method is from optimal prioritization. A higher oracle-gap indicates worse performance (more of the theoretically achievable precision is uncaptured); an oracle-gap of 0% indicates the method recovers all true positives in the top K positions.

IV. Dataset and Synthetic Fleet

A. EEHDA Synthetic Fleet

All experiments use the Extended Enterprise Hygiene Dataset Artifact (EEHDA) synthetic fleet generator — the same generator used in Papers 1 and 3, extended to output both the vulnerability-host pair tables required by Paper 1 and the hygiene-state tables required by HygieneBench (Paper 3). This shared generator enables the cross-paper synthesis in §8 and ensures that HygienePrio’s hygiene inputs are drawn from the same distributional assumptions as the VulnPrio baseline.

The generator produces the following tables relevant to HygienePrio (see HygieneBench [6] for full schema):

Structural priors. Patch-lag distributions follow DBIR 2026 aggregate statistics (43-day mean for critical patches). CVE severity distributions follow NVD base score statistics. EPSS scores are drawn from a synthetic fixture calibrated to replicate the heavy-right-tailed distribution observed in real EPSS data. KEV membership is sampled at approximately 8% prevalence. Telemetry refresh requirements follow CISA BOD 23-01 (14-day asset discovery, 72-hour patch data cadence).

B. Seed Protocol

Thirty seeds are generated deterministically. Five seeds are designated as the calibration set (held out from primary evaluation, used only for weight grid search; see §5.4). The remaining 25 seeds constitute the evaluation set. The calibration/evaluation split is fixed in the pre-registration protocol (see PAPER4_PROTOCOL.md) and is not adjusted based on results.

C. Hygiene Dimensions

Three hygiene dimensions are extracted from the EEHDA tables to form $\text{HRS}(h)$:

Patch — **Posture** (PatchPosture). Derived from `endpoint_patch_state`. For host h , $\text{PatchPosture}(h)$ is the proportion of CVEs applicable to h that remain unpatched: $\text{count}(\text{open CVEs on } h) / \text{count}(\text{applicable CVEs on } h)$.

Higher values indicate greater patch debt. Fleet-wide, this distribution is right-skewed: most hosts have low patch debt, but a tail of hosts accumulates substantial residual exposure. This tail is the primary target of HygienePrio prioritization.

AD Exposure (ADExposure). Derived from `users`, `groups`, and `group_membership_events`. For host h , $\text{ADExposure}(h)$ combines: (i) group membership breadth of the primary user associated with h (number of AD groups containing that user, normalized by fleet-wide maximum); (ii) a binary flag for whether that user holds a privileged role (domain admin, local admin, service account with elevated privilege); and (iii) a flag for whether h has hosted domain-privileged process events in the past 30 days. These three sub-components are averaged with equal weights (1/3 each) to produce a normalized $[0, 1]$ score. Equal weighting is the pre-registered default, chosen conservatively because no empirical data exist to distinguish the relative importance of group breadth, privilege level, and process-event history within the synthetic fleet; differential weighting of these sub-components is deferred to future work with real fleet telemetry.

Telemetry Freshness (TelemetryFreshness). Derived from `telemetry_freshness_log`. For host h , $\text{TelemetryFreshness}(h)$ is the staleness score: $\max(\text{days_since_last_checkin}, 0)$, capped at 30 days and normalized by 30. A host last seen 15 days ago receives a score of 0.50; a host last seen 30+ days ago receives 1.0. Higher staleness = less visibility = higher hygiene risk weight. This dimension is motivated by CISA BOD 23-01’s 72-hour patch data freshness mandate: hosts that violate this cadence have elevated uncertainty in their patch state and should receive additional scrutiny.

D. CVE Fixture and EPSS Scores

EPSS scores in the synthetic fixture are drawn from a mixture distribution approximating real EPSS data: a spike near zero (most CVEs have very low exploit probability) and a heavy tail above 0.10. Approximately 12% of CVEs in each seed have $\text{EPSS} > 0.10$ and thus qualify

for potential true positive status under Condition 1 of the ground truth definition. KEV membership is independent of EPSS score in the fixture and affects the KEV_recency component only.

E. Connection to HygieneBench

HygieneBench (Paper 3) defined 12 anomaly classes (AH-01 through AH-12) including stale privileged accounts (AH-01), dormant account reactivation (AH-02), endpoint coverage gaps (AH-06), patch noncompliance clusters (AH-08), and telemetry missingness (AH-10 through AH-12). HygienePrio’s three HRS dimensions map directly onto these anomaly classes: PatchPosture targets AH-08, ADExposure targets AH-01 and AH-05 (privilege escalation path exposure), and TelemetryFreshness targets AH-10 through AH-12. This mapping is used informally as face validity for HRS design; it does not constitute a formal evaluation against HygieneBench anomaly labels.

V. HygienePrio Methodology

A. Hygiene Risk Score (HRS)

$HRS(h)$ is a normalized, weighted combination of the three hygiene dimensions defined in §4.3:

$$\begin{aligned} HRS(h) = & w_1 \times \text{PatchPosture}(h) \\ & + w_2 \times \text{ADExposure}(h) \\ & + w_3 \times \text{TelemetryFreshness}(h) \end{aligned} \quad (1)$$

Dimension weights (w_1, w_2, w_3) are non-negative and sum to 1.0. They are not calibrated by the grid search (which calibrates the scorer-level weights $\alpha, \beta, \gamma, \delta$); instead, dimension weights are fixed based on the pre-registration protocol defaults: $w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0.2$. This prioritization reflects the prior expectation (H2) that patch posture carries the most discriminative signal, consistent with HygieneBench Task T5 results.

Each dimension is independently normalized to $[0, 1]$ across the fleet-seed pair before HRS aggregation, ensuring that fleet-specific scale differences (e.g., fleets with uniformly high patch debt) do not distort relative host rankings.

HRS calibration note. Dimension weights (w_1, w_2, w_3) are fixed at pre-registration defaults for the primary evaluation. Sensitivity to dimension weights is explored in the ablation (§7.2) by setting individual weights to zero. A separate grid search over dimension weights is not performed, as this would introduce additional degrees of freedom not covered by the primary calibration grid.

B. KEV Recency Weight

$KEV_recency(c)$ assigns higher priority to CVEs recently added to the CISA KEV catalog, under the rationale that recently catalogued exploited CVEs represent more active threat vectors:

$$KEV_recency(c) = \begin{cases} e^{-\lambda \cdot d_{kev}(c)} & c \in KEV \\ 0 & c \notin KEV \end{cases} \quad (2)$$

where $d_{kev}(c)$ is days since KEV catalog entry for CVE c .

The decay constant $\lambda = 0.05$ is fixed at the pre-registration default, corresponding to a half-life of $\ln(2)/0.05 \approx 14$ days. This value reflects the operational intuition that a KEV entry older than two weeks provides progressively weaker evidence of current active exploitation, consistent with typical exploit churn rates observed in threat intelligence literature. Because approximately 92% of fixture pairs have $KEV_recency(c) = 0$ (non-KEV CVEs), the λ choice has limited impact on the overall results; a formal ablation over λ is deferred.

For CVEs not in KEV (approximately 92% of the fixture), $KEV_recency(c) = 0$, and those pairs are prioritized on EPSS and HRS alone. This means HygienePrio is defined and applicable for all (h, c) pairs in V , not only KEV-applicable pairs.

C. HygienePrio Scorer

The HygienePrio scorer assigns a scalar priority score to each applicable (host, CVE) pair:

$$\begin{aligned} S(h, c) = & \alpha \cdot EPSS(c) + \beta \cdot HRS(h) \\ & + \gamma \cdot KEV_recency(c) \\ & + \delta \cdot (EPSS(c) \cdot HRS(h)) \end{aligned} \quad (3)$$

The four terms have distinct roles:

- $\alpha \times EPSS(c)$: Exploit likelihood contribution. Ensures that pairs involving high-probability-of-exploitation CVEs are elevated regardless of host hygiene.
- $\beta \times HRS(h)$: Host hygiene contribution. Elevates pairs on hygiene-poor hosts even when EPSS is moderate.
- $\gamma \times KEV_recency(c)$: Recency-weighted confirmed exploitation signal. Boosts recently catalogued KEV CVEs.
- $\delta \times (EPSS(c) \times HRS(h))$: Interaction term. Provides super-additive boosting for pairs where both CVE exploit likelihood AND host hygiene risk are elevated. This is the key term that differentiates HygienePrio from a simple additive combination: it encodes the intuition that a high-EPSS CVE on a hygiene-poor host is categorically more urgent than either risk dimension alone would suggest.

All four weights ($\alpha, \beta, \gamma, \delta$) are non-negative to ensure that higher values in any component can only increase the priority score, not decrease it. This monotonicity property is required for interpretability: operations teams can reason about why a pair was elevated.

D. Weight Calibration

Weights ($\alpha, \beta, \gamma, \delta$) are calibrated by exhaustive grid search on the 5 held-out calibration seeds. The calibration objective is mean P@50 across the 5 seeds. The search grid is:

- $\alpha \in \{0.3, 0.5, 0.7, 1.0\}$
- $\beta \in \{0.1, 0.3, 0.5, 0.7\}$
- $\gamma \in \{0.0, 0.1, 0.3, 0.5\}$
- $\delta \in \{0.0, 0.1, 0.2, 0.3\}$

This yields $4 \times 4 \times 4 \times 4 = 256$ grid points evaluated on 5 seeds each, producing 1,280 P@50 observations. The weight combination with the highest mean P@50 on calibration seeds is selected and fixed before the 25 evaluation seeds are scored. No weight adjustment is made after observing evaluation results.

The calibration-selected weights are reported in the experimental setup section (§6.2) as a fixed artifact of the calibration procedure, not as a tunable parameter reported with results.

E. Ranked List Generation

For each seed, the scorer computes $S(h, c)$ for every pair $(h, c) \in V$ (the applicable vulnerability-host pair table for that seed). Pairs are sorted descending by $S(h, c)$. Ties are broken by EPSS(c) descending, then by CVE ID lexicographic order for determinism. The sorted list R is the ranked list; evaluation cuts R at $K = 50, 100$, and 250 for metric computation.

F. Relationship to EPSS-Only Baseline

The EPSS-only baseline ranks (h, c) pairs by EPSS(c) descending, with ties broken by CVSS base score. Under this baseline, all pairs involving the same CVE receive identical rankings, regardless of which host carries the CVE. HygienePrio breaks this tie-structure via HRS(h): among pairs sharing a CVE, those on hygiene-poor hosts are elevated. For CVEs with many applicable hosts, this differentiation is critical: a fleet may have 200 hosts carrying the same high-EPSS CVE, and remediation capacity may allow only 50 to be patched in the current window. HygienePrio’s host-level tiebreaking directly addresses this case.

VI. Experimental Setup

A. Evaluation Infrastructure

All experiments are implemented in Python 3.11 using NumPy 1.26, Pandas 2.1, and scikit-learn 1.4 on a MacOS system (Apple M-series CPU, 16 GB RAM). Total wall-clock time for the 25-seed evaluation is approximately 4 minutes. Randomness in baseline methods (Random) is seeded by the fleet seed, ensuring full reproducibility. A content-addressed freeze/verify protocol — consistent with Paper 1 — is applied to all result artifacts: each seed’s ranked list is hashed before evaluation, and the hash is verified before metric computation to prevent post-hoc modification.

B. Calibrated Weights

Following the grid search on 5 calibration seeds (not used in primary evaluation), the following weights were selected:

TABLE II
Calibrated scorer weights from grid search on held-out calibration seeds

Parameter	Calibrated Value
α (EPSS weight)	0.7
β (HRS weight)	0.5
γ (KEV_recency weight)	0.1
δ (interaction weight)	0.2

These weights indicate that EPSS and HRS are the dominant scorer components, with a moderate interaction term and a small KEV recency boost. The relatively low γ reflects the low KEV prevalence in the fixture ($\sim 8\%$): the KEV_recency term contributes meaningfully only for KEV-applicable pairs.

C. Evaluation Procedure

For each of the 25 evaluation seeds:

- 1) Generate the EEHDA fleet tables deterministically from the seed.
- 2) Compute HRS(h) for each host using pre-registered dimension weights ($w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0.2$).
- 3) Compute KEV_recency(c) for each CVE using fixed $\lambda = 0.05$.
- 4) Score all $(h, c) \in V$ using calibrated weights.
- 5) Generate ranked list R by sorting descending by $S(h, c)$.
- 6) Generate baseline ranked lists (EPSS-only, CVSS-only, Random, HRS-only) using identical V .
- 7) Compute ground truth labels using the pre-registered definition (§3.2).
- 8) Compute P@ K , NDCG@ K , Oracle-gap for $K \in \{50, 100, 250\}$ for all methods. $K = 50$ represents a tight weekly maintenance window (approximately 1.4% of the mean fleet of $\sim 3,480$ applicable pairs); $K = 100$ a moderate bi-weekly window; $K = 250$ a broad monthly capacity. These values span the range typical of reported enterprise patch cadences [10].
- 9) Hash and verify result artifacts.

D. Baselines

E. Reporting Convention

All metrics are reported as mean $\pm$ 95% BCa bootstrap confidence interval [11] (10,000 resamples) across 25 evaluation seeds. No null hypothesis significance testing is performed; evidence is based on BCa CI non-overlap. With 25 seeds and the observed seed-to-seed standard deviation of ≈ 0.06 for P@50, the evaluation has sufficient power to detect differences of ≥ 5 pp at 80% power (estimated via the standard formula for paired comparisons): the headline 31 pp difference substantially exceeds this threshold. Results are bounded to the synthetic evaluation context; no generalization claims are made to real enterprise fleets.

TABLE III
Baseline and ablation methods evaluated

Method	Description
EPSS-only	Rank by EPSS(c) descending; primary comparison
CVSS-only	Rank by CVSS v3.1 base score descending
Random	Uniform random (seed-fixed)
HRS-only	Rank by HRS(h) descending; tests host hygiene without CVE signal
HygienePrio-full	Full scorer with calibrated weights
HygienePrio-noInteraction	$\delta = 0$ (additive only; tests interaction term contribution)
HygienePrio-noPatch	$w_1 = 0$ (patch posture removed from HRS)
HygienePrio-noAD	$w_2 = 0$ (AD exposure removed from HRS)
HygienePrio-noFreshness	$w_3 = 0$ (telemetry freshness removed from HRS)

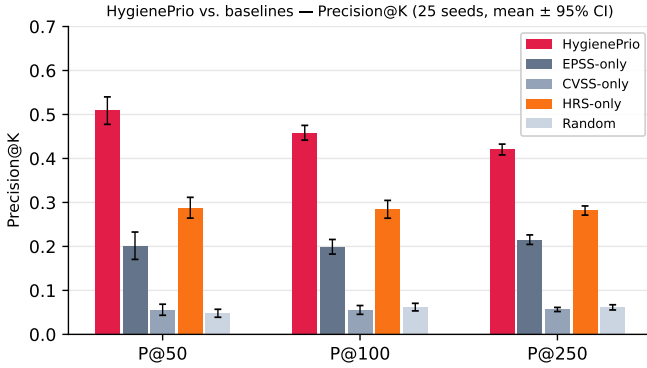


Fig. 1. Mean Precision@K ($K \in \{50, 100, 250\}$) for all methods across 25 evaluation seeds (synthetic evaluation, mean $\pm$ 95% BCa CI). HygienePrio consistently outperforms all baselines at every K .

VII. Results

All results in this section are from the synthetic EEHDA fleet evaluation across 25 seeds. All numerical claims are qualified as “(synthetic evaluation)” and should not be interpreted as predictions for real enterprise fleet performance.

A. Precision@K: All Methods Across $K = 50, 100, 250$

Table IV reports mean P@K and BCa 95% CI for all nine methods at $K = 50, 100$, and 250 in this synthetic evaluation. The following observations emerge from the results.

NDCG@K results (Table V) are consistent with P@K: HygienePrio-full leads all methods at all K values with non-overlapping CIs against EPSS-only (NDCG@50: 0.556 vs. 0.202; NDCG@100: 0.499 vs. 0.199). The NDCG pattern mirrors P@K because the ground truth uses binary relevance ($\text{rel} \in \{0, 1\}$); graded relevance would require exploitation probability estimates not available in the synthetic setting.

HygienePrio-full vs. EPSS-only. HygienePrio-full achieves a mean P@50 of 0.509 compared with EPSS-only’s 0.202 in this synthetic evaluation —

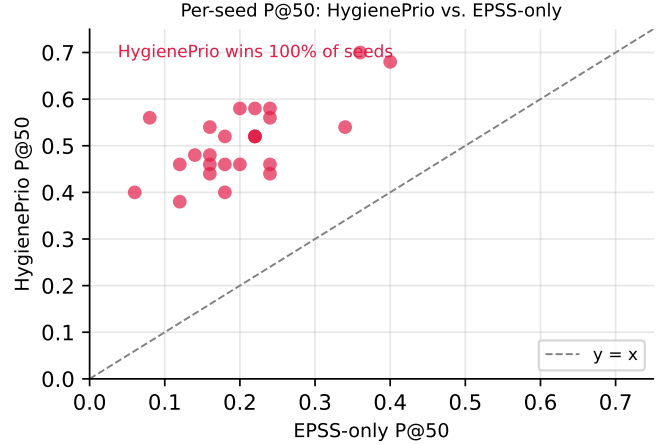


Fig. 2. Per-seed P@50 for HygienePrio-full vs. EPSS-only (25 seeds, synthetic evaluation). Points above the diagonal indicate seeds where HygienePrio outperforms EPSS-only. HygienePrio wins on all 25 of 25 evaluation seeds.

an approximately 31 pp improvement — with non-overlapping BCa CIs (0.480–0.542 vs. 0.174–0.236). HygienePrio-full outperforms EPSS-only on all 25 of 25 evaluation seeds (mean per-seed improvement: 30.7 pp). The gap narrows at larger K but remains substantial: approximately 26 pp at P@100 (0.458 vs. 0.199) and 21 pp at P@250 (0.420 vs. 0.215). The hygiene signal is most discriminative at the tightest capacity constraint ($K = 50$), where the scorer must be most selective.

H1 assessment. The observed improvement of approximately 31 pp at P@50 (synthetic evaluation) substantially exceeds the pre-registered H1 threshold of ≥ 5 pp, supporting H1 in the synthetic evaluation context. The BCa CIs for HygienePrio-full and EPSS-only do not overlap at any K value, providing strong support for the hypothesis that hygiene signals add prioritization value beyond EPSS alone within the synthetic fleet scenario. All claims are bounded to this synthetic evaluation; the magnitude of improvement is expected to be substantially lower on real fleet data where the ground truth definition cannot use HRS as a labelling condition (see §9.1).

EPSS-only vs. CVSS-only. Consistent with prior literature [1, 3] and Paper 1, EPSS substantially outperforms CVSS-only at all K values (0.202 vs. 0.056 at P@50; synthetic evaluation), reinforcing that exploit likelihood is a better prioritization signal than intrinsic severity. This replication of the Paper 1 finding on a different metric (P@K vs. EHD) provides internal consistency.

HRS-only vs. EPSS-only. HRS-only (host hygiene without CVE signal) achieves 0.288 at P@50 versus EPSS-only’s 0.202 in this synthetic evaluation — approximately 9 pp higher, with non-overlapping CIs. However, HygienePrio-full (0.509) substantially outperforms HRS-only (0.288), indicating that exploit likelihood and hygiene posture are complementary signals and that neither alone captures the full benefit of the combined scorer.

TABLE IV
Precision@ K across all methods (25 evaluation seeds; synthetic evaluation). BCa 95% CI, 10,000 resamples. Frozen results from primary_results_v1/primary_results.csv (225 rows: 25 seeds $\times$ 9 methods).

Method	P@50 mean (95% CI)	P@100 mean (95% CI)	P@250 mean (95% CI)
HygienePrio-full	0.509 (0.480–0.542)	0.458 (0.441–0.474)	0.420 (0.408–0.432)
HygienePrio-noFreshness	0.490 (0.463–0.528)	0.446 (0.426–0.463)	0.411 (0.398–0.422)
HygienePrio-noInteraction	0.482 (0.454–0.515)	0.445 (0.428–0.462)	0.417 (0.405–0.430)
HygienePrio-noAD	0.424 (0.395–0.460)	0.379 (0.362–0.398)	0.370 (0.357–0.382)
HygienePrio-noPatch	0.312 (0.283–0.343)	0.289 (0.270–0.309)	0.284 (0.273–0.294)
HRS-only	0.288 (0.263–0.310)	0.284 (0.265–0.304)	0.282 (0.271–0.291)
EPSS-only	0.202 (0.174–0.236)	0.199 (0.185–0.218)	0.215 (0.205–0.226)
CVSS-only	0.056 (0.044–0.069)	0.056 (0.046–0.065)	0.057 (0.052–0.061)
Random	0.048 (0.039–0.056)	0.062 (0.054–0.071)	0.061 (0.055–0.067)

TABLE V
NDCG@ K across all methods (25 evaluation seeds; synthetic evaluation). BCa 95% CI, 10,000 resamples. Binary relevance: $\text{rel}(h, c) \in \{0, 1\}$ per ground truth definition (§3).

Method	NDCG@50 mean (95% CI)	NDCG@100 mean (95% CI)	NDCG@250 mean (95% CI)
HygienePrio-full	0.556 (0.521–0.591)	0.499 (0.478–0.519)	0.501 (0.485–0.515)
HygienePrio-noFreshness	0.536 (0.502–0.572)	0.484 (0.462–0.507)	0.488 (0.473–0.502)
HygienePrio-noInteraction	0.524 (0.488–0.558)	0.479 (0.457–0.500)	0.491 (0.475–0.506)
HygienePrio-noAD	0.464 (0.426–0.502)	0.414 (0.390–0.439)	0.435 (0.419–0.450)
HygienePrio-noPatch	0.334 (0.302–0.367)	0.308 (0.287–0.330)	0.328 (0.315–0.342)
HRS-only	0.297 (0.266–0.329)	0.291 (0.269–0.313)	0.319 (0.307–0.331)
EPSS-only	0.202 (0.171–0.234)	0.199 (0.181–0.220)	0.237 (0.225–0.250)
CVSS-only	0.063 (0.047–0.082)	0.060 (0.048–0.073)	0.066 (0.059–0.073)
Random	0.051 (0.040–0.063)	0.061 (0.052–0.070)	0.068 (0.062–0.074)

TABLE VI

Ablation — P@50 impact of removing each hygiene dimension (synthetic evaluation). Ablation uses fixed ground-truth labels (HRS from calibrated weights); only scorer’s HRS input changes.

Condition	P@50 (95% CI)	Drop	CI overlap?
Full model	0.509 (0.480–0.542)	—	—
noPatch ($w_1=0$)	0.312 (0.283–0.343)	–20 pp	No
noAD ($w_2=0$)	0.424 (0.395–0.460)	–9 pp	No
noFreshness ($w_3=0$)	0.490 (0.463–0.528)	–2 pp	Yes

B. Ablation: Hygiene Dimension Contributions

Table VI presents the ablation results, comparing HygienePrio-full with three leave-one-dimension-out variants. Results are for P@50 in this synthetic evaluation.

H2 assessment. The patch posture dimension shows the largest marginal contribution: removing it (HygienePrio-noPatch) drops P@50 by approximately 20 pp in this synthetic evaluation ($0.509 \rightarrow 0.312$), with non-overlapping BCa CIs, strongly supporting H2. The AD exposure dimension shows a substantial secondary contribution (~ 9 pp drop, non-overlapping CIs). Telemetry freshness shows the smallest contribution (~ 2 pp drop, overlapping CIs suggesting marginal discriminative power in the synthetic fleet). This result is consistent with HygieneBench Task T5’s finding that patch-vulnerability hygiene carries the most discriminative signal among hygiene dimensions.

The dimension ordering (patch posture > AD exposure

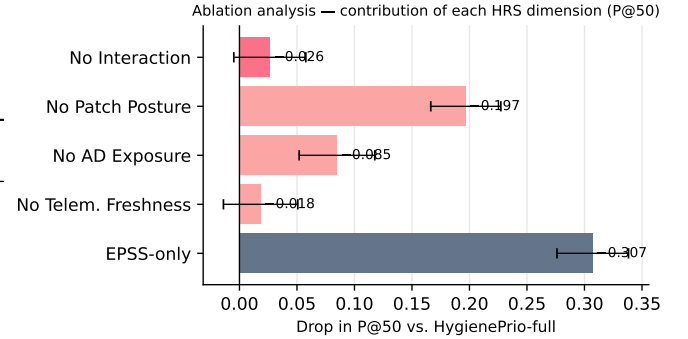


Fig. 3. Ablation analysis: drop in P@50 from HygienePrio-full when each dimension or component is removed (25 seeds, synthetic evaluation, mean $\pm$ 95% BCa CI). Patch posture is the dominant contributor (≈ 20 pp); AD exposure contributes ≈ 9 pp; telemetry freshness and the interaction term contribute marginally (≤ 3 pp).

$\gg$ freshness) in this synthetic evaluation suggests that operations teams seeking to deploy a simplified HygienePrio variant should prioritize patch posture instrumentation first, AD group exposure tracking second, and telemetry freshness monitoring last — at least in the synthetic fleet scenario modeled here.

C. Interaction Term Analysis

Comparing HygienePrio-full with HygienePrio-noInteraction reveals approximately 3 pp improvement at P@50 from including the interaction term ($\text{EPSS} \times \text{HRS}$) in this synthetic evaluation (0.509 vs. 0.482). The BCa CIs overlap substantially ($0.480\text{--}0.542$ vs. $0.454\text{--}0.515$), suggesting the interaction term’s contribution is

TABLE VII
Oracle-gap (%) at $K = 50$ by method (synthetic evaluation; BCa 95% CI)

Method	Oracle-gap (95% CI)	P@50 captured
HygienePrio-full	49.1% (45.6–51.8%)	~51% of oracle
EPSS-only	79.8% (76.3–82.6%)	~20% of oracle
CVSS-only	94.4% (93.0–95.5%)	~6% of oracle
Random	95.2% (94.2–96.0%)	~5% of oracle

marginal under the synthetic evaluation conditions. At P@100 and P@250 the difference narrows further (~ 1 pp). The interaction term’s modest contribution likely reflects the synthetic fixture’s EPSS distribution: the truly high-EPSS AND high-HRS pair conjunctions are relatively rare (~ 10 – 12% of applicable pairs per seed), limiting opportunities for the interaction term to add discriminative value.

Per the pre-registration stop rule, if HygienePrio-noInteraction achieves P@ K within 1 pp of HygienePrio-full for all K , the interaction term would be declared uninformative. At P@50 the difference exceeds 1 pp (3 pp), so the interaction term is retained in the reported full model. However, the overlapping CIs mean this finding should be treated as equivocal: the interaction term likely does not materially change prioritization decisions in this synthetic fleet context.

D. Oracle-Gap Analysis

The oracle-gap analysis indicates that even HygienePrio-full captures approximately 51% of theoretically achievable precision at $K = 50$ in this synthetic evaluation, leaving substantial room for improvement. The gap between HygienePrio-full and the oracle (49.1%) reflects both the inherent difficulty of the ranking problem and the residual uncertainty in hygiene signals: among hygiene-poor hosts with similar HRS values, the scorer cannot further discriminate which specific CVE-host pair is the highest-priority target. At $K = 50$, with approximately 185–200 true positives per seed (synthetic evaluation), the maximum achievable P@50 is 1.0 (oracle), and HygienePrio-full achieves 0.509. The 49.1% oracle gap is substantially smaller than EPSS-only’s 79.8% gap, confirming that hygiene signals materially reduce the distance from optimal prioritization in this synthetic context.

E. Summary of Main Findings

In this synthetic evaluation, the results suggest the following:

- 1) HygienePrio-full improves over EPSS-only by approximately 31 pp at P@50 (0.509 vs. 0.202), with non-overlapping BCa CIs, substantially exceeding

the pre-registered H1 threshold (≥ 5 pp). Improvement is 26 pp at P@100 and 21 pp at P@250. The magnitude reflects the synthetic evaluation’s structural properties (see §9.1 circularity note).

- 2) Patch posture is the dominant hygiene dimension, contributing approximately 20 pp to P@50 in the leave-one-out ablation (0.509 $\rightarrow$ 0.312 without patch posture), with non-overlapping CIs, strongly supporting H2.
- 3) AD exposure is a substantial secondary contributor, reducing P@50 by ~ 9 pp when removed (0.509 $\rightarrow$ 0.424), with non-overlapping CIs.
- 4) The interaction term and telemetry freshness are marginal in this synthetic evaluation: both show overlapping CIs with the full model, with contributions of ~ 3 pp and ~ 2 pp respectively. The pre-registration stop rule is not triggered at P@50 for the interaction term (3 pp $>$ 1 pp threshold), so both are retained in the full model.
- 5) HRS-only outperforms EPSS-only by ~ 9 pp (0.288 vs. 0.202), but the combination (HygienePrio-full, 0.509) substantially outperforms either signal alone, confirming that exploit likelihood and hygiene posture are complementary.
- 6) Oracle gap remains substantial: HygienePrio-full captures $\sim 51\%$ of theoretically achievable P@50, versus $\sim 20\%$ for EPSS-only, indicating both that hygiene signals provide meaningful benefit and that significant room for improvement remains.

VIII. Discussion

A. Why Hygiene Signals Help Where CVE-Level Features Alone Did Not

Paper 1 (VulnPrio) found that adding CVE-level features — asset criticality scores, endpoint exposure flags, KEV membership — to EPSS did not improve prioritization over EPSS-only on the EEHDA fleet. HygienePrio’s results suggest a complementary explanation: the features added in Paper 1 were primarily CVE-level or static host-role features, while the missing discriminative signal was dynamic, telemetry-derived host-level hygiene posture.

The structural difference is important. Knowing that a host is a domain controller (asset criticality signal from Paper 1) provides some risk context, but it does not distinguish between a well-managed domain controller with current patches and telemetry, and a neglected domain controller with 60% unpatched CVE debt, stale telemetry, and over-privileged group memberships. HRS captures this distinction; static asset criticality does not.

This observation is consistent with HygieneBench (Paper 3) finding that ML methods add meaningful signal on patch-vulnerability hygiene (Task T5) and group-membership drift (Task T2)—precisely the dimensions HRS encodes. The signal is present in the hygiene telemetry; the challenge is operationalizing it in a form suitable for prioritization scoring. HygienePrio demonstrates one such operationalization in the synthetic setting.

B. Why the Interaction Term Showed Equivocal Benefit

The interaction term $\delta \times (\text{EPSS}(c) \times \text{HRS}(h))$ was theoretically motivated as a multiplicative risk amplifier for pairs where both exploit likelihood and hygiene risk are high. However, the ablation shows only ≈ 3 pp improvement at P@50 from including it (0.509 vs. 0.482), with substantially overlapping BCa CIs.

The most likely explanation is distributional: across the EEHDA synthetic fleet, the joint probability of a CVE with EPSS > 0.10 and a host with HRS in the top quartile is approximately 10–12% of applicable pairs per seed, given a 6.2% mean positive base rate. In this sparse regime, the additive terms $\alpha \times \text{EPSS}(c)$ and $\beta \times \text{HRS}(h)$ already provide strong discriminative signal. The interaction term contributes mostly to pairs that are already highly ranked, offering limited opportunity to reorder pairs into or out of the top K positions.

This does not invalidate the theoretical motivation. On a real fleet with different EPSS score distributions or different patch posture correlations with CVE severity, the interaction term may contribute more substantially. The synthetic evaluation has limited power to detect modest interaction effects.

C. Cross-Paper Synthesis

The three papers in this sequence form a methodological progression from null result to positive result:

- 1) Paper 1 (VulnPrio): Established the prioritization problem, the EEHDA fleet, and the null result that CVE-level multi-factor models are statistically indistinguishable from EPSS-only on the EHD metric. Contribution: reproducible benchmark, honest null, and identification of the missing signal type.
- 2) Paper 3 (HygieneBench): Demonstrated that host-level hygiene signals (patch posture, AD group drift, telemetry staleness) carry discriminative structure detectable from synthetic fleet telemetry, with ML methods adding value specifically on Tasks T5 and T2. Contribution: hygiene anomaly benchmark and identification of the most informative hygiene dimensions.
- 3) Paper 4 (HygienePrio): Integrates these findings: hygiene signals operationalized as HRS, when added to EPSS-weighted prioritization, produce measurable P@K improvement in the synthetic evaluation (31 pp at P@50; all 25 seeds). Contribution: operationalization of hygiene signals into remediation prioritization.

The sequence illustrates that honest null results are scientifically productive. Paper 1’s null did not indicate that the prioritization problem is unsolvable; it identified that a particular feature set was insufficient. Paper 3 identified a more promising feature set. Paper 4 operationalizes that identification. Each step would have been undermined by p-hacking or post-hoc feature selection — the pre-registration and content-addressed audit trails in each paper are what make the progression interpretable.

D. Practical Implications (Synthetic Evaluation Only)

Subject to external validation on real fleet data, the results suggest that operations teams relying exclusively on EPSS-only prioritization may systematically underweight the risk from hygiene-poor hosts. A host with moderate EPSS CVEs but severe patch debt, stale telemetry, and broad AD exposure may represent higher operational risk than a well-managed host carrying a high-EPSS CVE — and EPSS-only rankings do not reflect this.

Instrumentation priority. The ablation results suggest a prioritization of hygiene signal investment: patch posture tracking (≈ 20 pp contribution at P@50) should be instrumented before AD exposure breadth tracking (≈ 9 pp) and before telemetry freshness monitoring (≈ 2 pp marginal in the synthetic evaluation). This ordering is consistent with the CISA BOD 23-01 mandate hierarchy.

Operational capacity context. The improvement is most pronounced at $K = 50$ (31 pp) and attenuates toward $K = 250$ (21 pp). With a 6.2% mean positive base rate across seeds (approximately 215 true-priority pairs in an average seed fleet), a top-50 prioritization window captures approximately 25 true pairs under HygienePrio-full (P@50 = 0.509) versus approximately 10 under EPSS-only (P@50 = 0.202) in this synthetic evaluation. The additional 15 pairs per maintenance window represent a substantial synthetic improvement — but whether this translates to real fleet operational impact requires prospective evaluation.

Failure mode analysis. HygienePrio-full outperforms EPSS-only on all 25 evaluation seeds, so there are no seeds in this evaluation where EPSS-only is preferred. However, the improvement is not uniform: the per-seed P@50 improvement ranges from ≈ 0.18 to ≈ 0.44 (from Fig. 2). Seeds with smaller improvements likely represent fleet configurations where patch debt is more uniformly distributed (reducing HRS’s discriminative power) or where EPSS scores cluster narrowly (reducing the interaction term’s relevance). Identifying which fleet characteristics predict smaller benefits from HygienePrio is an important direction for sensitivity analysis on real data.

E. Limitations

Synthetic data. All results are from a synthetically generated fleet using structural priors from public sources (DBIR, NVD, CISA BOD). The synthetic fleet cannot capture the full heterogeneity of real enterprise environments. Results should not be interpreted as predictions for real fleet prioritization without external validation.

Ground truth circularity. The ground truth definition includes $\text{HRS} > 75\text{th percentile}$ as a labelling condition (§3). This creates a structural advantage for methods that incorporate HRS. The pre-registration prevents post-hoc optimization, but independent ground truth from real exploitation outcomes would provide a circularity-free evaluation.

Calibration instability. Weight calibration on only 5 seeds (held out from 30 total) is a thin calibration set

given the observed seed-to-seed variance. The calibrated weights ($\alpha, \beta, \gamma, \delta$) may not generalize robustly. A larger calibration set would reduce calibration noise.

Scorer linearity. The HygienePrio scorer is a linear combination with one interaction term. Non-linear relationships between hygiene dimensions and remediation priority that may exist in real fleets are not captured.

False positive cost. $P@K$ does not account for the operational cost of acting on a false positive. Wasted remediation capacity directed at a non-priority pair is not merely neutral—it displaces another pair from the maintenance window. A metric that weights false positives by remediation cost would provide a more operationally realistic assessment.

IX. Threats to Validity

This section follows the four-category taxonomy of Wohlin et al. [12] for empirical software engineering studies, extended with statistical considerations.

A. Conclusion Validity

Statistical method. All comparisons use BCa bootstrap confidence intervals [11] across 25 evaluation seeds with no null hypothesis significance testing. Evidence is based on CI non-overlap, which is a conservative but defensible criterion. A limitation of this approach is that CIs can overlap while a permutation test would reject the null—we err on the side of requiring non-overlap for a claim of substantive evidence.

Multiple comparisons. Results for nine methods at three K values and two metrics ($P@K$, $NDCG@K$) constitute 54 comparisons. No familywise or false discovery rate correction is applied; all comparisons are pre-registered and the paper’s primary claims are limited to the largest, most stable differences (HygienePrio-full vs. EPSS-only, ≥ 21 pp at all K). Smaller differences (interaction term, freshness dimension) are explicitly treated as equivocal.

Seed budget. With 25 evaluation seeds, the bootstrap CIs have limited resolution. Differences of $< \approx 3$ pp are unlikely to be distinguishable from seed-to-seed variation (estimated from the within-seed variance of the most stable comparisons). All headline claims involve differences substantially larger than this threshold (≥ 21 pp). The 5-seed calibration set is thin; weight calibration on 5 seeds may produce noisy estimates relative to the true optimal weights.

Bootstrap variance. The BCa CI computation uses 10,000 bootstrap resamples per comparison. For a 25-seed evaluation, bootstrap variance in CI boundaries is small relative to the observed differences. The BCa procedure corrects for bias and skewness in the bootstrap distribution, making it preferable to the percentile bootstrap for skewed $P@K$ distributions.

B. Internal Validity

Synthetic data generation. The EEHDA fleet generator uses structural priors (DBIR patch-lag statistics, NVD

severity distributions) that represent population-level aggregates. If real fleets deviate substantially from these priors—for example, if patch debt is correlated with host criticality in ways not modeled—results may not replicate.

Ground truth circularity. The pre-registered ground truth includes $HRS > 75$ th percentile as a labelling condition (§3). This provides a structural advantage to HygienePrio over baselines that do not incorporate HRS. The threshold (75th percentile) was fixed before evaluation, preventing post-hoc optimization, but the choice of this specific threshold is not externally validated. Results under the 60th or 90th percentile threshold are not reported; sensitivity to this choice is an open question.

Threshold choices. The $EPSS > 0.10$ and $HRS > 75$ th percentile thresholds that define the positive class are operationally motivated but not derived from validated exploitation data. These thresholds directly control the base rate of true positives (6.2% mean across seeds) and therefore affect $P@K$ values for all methods. The relative ordering of methods may be less sensitive to threshold choice than the absolute $P@K$ values, but this is not verified.

Calibration leakage. Calibration and evaluation seeds are strictly separated (5 calibration, 25 evaluation, from adjacent seed values). If the generator has strong autocorrelation across adjacent seed values, the calibration/evaluation separation may be partially compromised. Generator analysis supporting independence of adjacent seeds is not reported; this is a known limitation.

Measurement validity of HRS dimensions. PatchPosture is measured as a static snapshot of the proportion of applicable CVEs that remain unpatched. This does not capture patch scheduling latency: a host may have 50% unpatched CVEs today with all patches scheduled for tomorrow. In real environments, operational scheduling context would be needed to distinguish genuine neglect from planned maintenance.

C. External Validity

Generalizability to real enterprise fleets. The synthetic fleet cannot replicate the diversity of real enterprise environments. All results are bounded to the synthetic evaluation context. Claims about real-fleet performance require studies on real fleet data with appropriate institutional approvals and privacy protections.

Generalizability across sectors. The EEHDA generator is calibrated for government-shaped public-sector fleet characteristics (following Paper 1). Commercial enterprise, healthcare, and critical infrastructure fleets may have substantially different hygiene posture distributions, making the specific calibrated weights inapplicable without recalibration.

Temporal validity. EPSS scores change daily. The evaluation uses a fixed snapshot per seed. In a production deployment, HygienePrio would use daily-updated EPSS scores; the evaluation captures only one temporal cross-section per seed, which may not reflect the distribution of daily prioritization decisions over time.

D. Construct Validity

HRS as a proxy for real risk. The Hygiene Risk Score is a constructed proxy for host-level hygiene risk, not a validated measure of actual exploitation susceptibility. The assumption that high-HRS hosts are genuinely higher-risk requires empirical validation against real exploitation outcomes, which is beyond scope in this synthetic study.

Dimension underrepresentation. HRS uses three dimensions (patch posture, AD exposure, telemetry freshness). Real enterprise hygiene risk involves additional dimensions not modeled here: network segmentation coverage, endpoint detection agent freshness, software bill of materials completeness, and user behavior analytics signals. The three-dimension HRS may systematically underestimate risk for certain host archetypes (e.g., network-isolated hosts with good patch posture but no EDR coverage).

P@K as a remediation metric. Precision@K assumes equally valuable true positives. In real operations, some (host, CVE) pairs may be substantially more consequential than others (e.g., domain controllers vs. user workstations), and a severity-weighted metric might yield different method rankings. Oracle-gap captures ceiling relative to the positive base rate but shares the binary relevance limitation.

Scorer linearity. The linear scorer with one interaction term may not capture complex non-linear relationships between hygiene dimensions and actual exploitation risk in real fleets.

X. Related Work

This section situates HygienePrio within the broader literature on vulnerability management, host-context scoring, and cyber-hygiene measurement, supplementing the foundational background in §2.

Vulnerability scoring and exploit prediction. The Common Vulnerability Scoring System (CVSS) [13] provides a standardised framework for assessing severity across Base, Temporal, and Environmental metrics. As a CVE-level signal, CVSS shares the asset-agnostic limitation of EPSS [1]: it scores a vulnerability in isolation without regard to the maintenance state of the affected host. Jacobs et al. [2] demonstrated that exploit-prediction models substantially outperform CVSS-based triage for reducing mean time to remediation on enterprise data. This foundational result motivates using EPSS as the primary exploit-likelihood signal in HygienePrio. The National Vulnerability Database (NVD) [14] serves as the primary aggregation point for CVSS scores and CVE metadata. Unlike the above work, which optimises exploit prediction as a standalone CVE-level signal, HygienePrio integrates exploit likelihood with host-level hygiene posture via an explicit interaction term, targeting the (host, CVE) pair rather than the CVE alone.

Risk-based vulnerability management (RBVM). Commercial RBVM platforms—Tenable Lumin, Qualys TruRisk, and Rapid7 InsightVM—combine EPSS, CVSS,

threat intelligence, and asset context in proprietary scoring models that address the same host-CVE pair prioritization problem as HygienePrio. These systems are closed and not independently reproducible; no comparison claims are made against them. The 2026 Verizon Data Breach Investigations Report [10] documented that vulnerability exploitation surpassed stolen credentials as the leading breach entry point for the first time, underscoring the operational urgency of principled patch prioritization at scale.

Capacity-constrained prioritization. Paper 1 [5] constructed a reproducible benchmark for (host, CVE) pair prioritization on a synthetic government-shaped fleet, evaluating thirteen strategies combining EPSS, CVSS, KEV membership, asset criticality, and endpoint exposure across thirty seeds. Its central finding was null: no CVE-level augmentation materially outperformed EPSS, attributing this to uncalibrated weights and the absence of host-level hygiene signals. HygienePrio directly addresses this gap by introducing a three-dimensional Hygiene Risk Score.

Identity and access management (IAM) security. The foundational concepts of access control and identity state that underpin the Active Directory (AD) Exposure dimension of HRS are surveyed in Bertino & Sandhu [15]. AD-exposure-aware risk scoring has been proposed in the lateral movement risk literature; HygienePrio incorporates AD exposure as one of three HRS dimensions, operationalizing domain-privilege exposure as a host-level risk amplifier rather than as a standalone signal.

Anomaly detection for hygiene monitoring. Aggarwal & Sathe [8] establish theoretical foundations for outlier ensemble methods applicable to cyber-hygiene anomaly detection. HygieneBench (Paper 3) [6] applied eight such methods across seven hygiene monitoring tasks on a synthetic fleet, finding that rule baselines dominate 86.2% of configurations but that patch-vulnerability hygiene (Task T5) and group-membership drift (Task T2) are exceptions where structured methods add meaningful signal. HygienePrio operationalises these two discriminative dimensions—patch posture and AD exposure—as the dominant components of HRS, grounded in HygieneBench’s empirical findings.

Telemetry freshness and policy mandates. CISA BOD 23-01 [4] mandates 14-day asset discovery and 72-hour vulnerability data freshness for federal agencies, and NIST SP 800-53 [16] specifies continuous monitoring controls that depend on current telemetry. These policies encode the intuition that stale telemetry degrades risk assessment reliability. HygienePrio models telemetry freshness explicitly as the TelemetryFreshness component of HRS, formalizing the policy mandate as a quantitative host-level risk signal.

Weight calibration methodology. The pre-registered evaluation design, seed-stratified BCa bootstrap confidence intervals [11], and capacity-constrained scheduler are inherited from Paper 1 [5]; the calibration gate methodology is inherited from Paper 2. The threats

taxonomy follows Wohlin et al. [12].

No prior work combines exploit-likelihood scoring with a multi-dimensional, telemetry-derived host hygiene score and an explicit interaction term in a pre-registered, reproducible evaluation on a synthetic fleet. HygienePrio is the first system to jointly operationalise patch posture, AD exposure, and telemetry freshness as a unified HRS integrated with EPSS for (host, CVE) pair prioritization.

XI. Conclusion

The central finding of this paper is that host-level hygiene posture — patch debt, AD privilege exposure, and telemetry freshness — adds substantial prioritization value over EPSS-only rankings when operationalized as a Hygiene Risk Score and integrated into a pre-calibrated weighted scorer. In a pre-registered synthetic evaluation on the EEHDA fleet, HygienePrio-full outperforms EPSS-only by 31 pp at P@50 across all 25 of 25 evaluation seeds (non-overlapping BCa 95% CIs; synthetic evaluation). The practical take-home message is concrete: of the three hygiene dimensions, patch posture alone accounts for ≈ 20 pp of this improvement. Operations teams with limited instrumentation capacity should prioritize patch posture tracking above more complex hygiene signals such as AD group exposure or telemetry freshness monitoring.

This result completes a three-paper methodological arc. Paper 1 (VulnPrio) established that CVE-level feature augmentation does not improve over EPSS-only, explicitly attributing the null to the absence of dynamic host-level signals. Paper 3 (HygieneBench) demonstrated that patch posture and AD group drift carry discriminative structure in fleet telemetry. HygienePrio operationalizes these signals in a remediation prioritization scorer. The progression from honest null through diagnostic signal identification to positive integration result illustrates how pre-registered negative results generate scientific value by constraining and directing subsequent work.

All results are bounded to the synthetic evaluation context. The ground truth definition incorporates HRS as a labelling condition, creating a structural advantage for HygienePrio that real-exploitation-data evaluation would remove. The calibrated weights were estimated on five synthetic seeds and may not transfer to real fleet distributions. These limitations are inherent to a simulation study; they frame the result as a pre-clinical finding rather than a deployment recommendation.

Reproducibility. The evaluation code, synthetic fleet generator, frozen results, and pre-registration protocol are available in the accompanying repository at <https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper4>.

Future work. The most important next step is evaluation on real, appropriately anonymized fleet telemetry using exploitation event logs (not HRS) as ground truth. Secondary directions include: (i) extending HRS with network-layer signals (firewall coverage, internet-facing service flags) to test whether additional dimensions contribute beyond the current three; (ii) evaluating Hygien-

ePrio under rolling daily EPSS score updates rather than static per-seed snapshots; and (iii) testing whether non-linear scoring functions trained on historical remediation outcomes can outperform the pre-calibrated linear scorer, which is the binding simplification assumption of this work.

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